

4 Hyperbolic Geometry

4.1 History: Saccheri, Lambert and Absolute Geometry

For 2000 years after Euclid, many mathematicians believed that his parallel postulate (page 12) could not be an independent axiom. Much effort was expended trying to demonstrate this, not least by Giovanni Saccheri (1667–1733) & Johann Lambert (1728–1777); both attempted to force contradictions by assuming the negation of the parallel postulate. While this approach ultimately failed, their insights provided the foundation of a new *non-Euclidean* geometry.

Absolute and Non-Euclidean Geometry

Before considering the work of Saccheri and Lambert, we define some terms and recall our earlier discussion of parallels (pages 13–17).

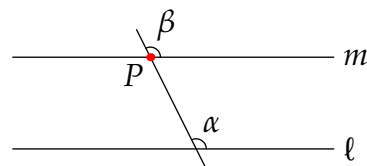
Definition 4.1. *Absolute or neutral geometry* is the axiomatic system comprising all of Hilbert's axioms except Playfair (page 21). Euclidean geometry is a special case of absolute geometry. A *non-Euclidean geometry* is (loosely) a model satisfying most of Hilbert's axioms but for which parallels might not exist or be unique:

There exists a line ℓ and a point P not on ℓ which lies on *no parallel* or *at least two parallel lines* to ℓ .

Spherical geometry (page 16) is non-Euclidean since there are no parallel lines: Hilbert's axioms I-2 and O-3 are false, as is the exterior angle theorem.

Results in absolute geometry The conclusions of Euclid's first 28 theorems are valid.

- Basic constructions: midpoint, bisectors, perpendiculars, etc.
- Triangle congruence theorems: SAS, ASA, SAA, SSS.
- Exterior angle theorem and its consequences:
 - Side-angle comparison and the triangle inequality (Exercise 2.3.6).
 - *Existence* of a parallel m to a line ℓ through a point $P \notin \ell$: as pictured $\alpha \cong \beta \implies \ell \parallel m$.



Arguments making use of unique parallels Since the following were proved using Playfair's axiom or the parallel postulate, their arguments are false in absolute geometry:

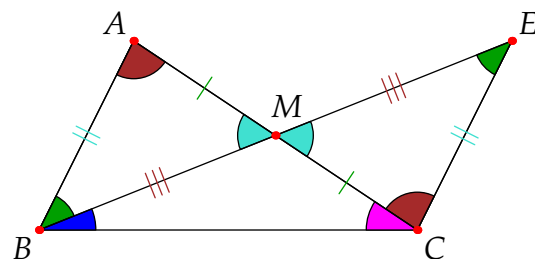
- A line crossing parallel lines makes congruent angles: in the picture, $\ell \parallel m \implies \alpha \cong \beta$. This is the *uniqueness* claim in Playfair: the parallel m to ℓ through P is unique.
- Angles in a triangle sum to a straight-edge (180°).
- Constructions of squares/rectangles.
- Pythagoras' Theorem.

While our *arguments* for the above are false in absolute geometry, we cannot immediately claim that the *results* are also; there might be alternative proofs! To see that these results truly require unique parallels, we must exhibit a *model* of absolute geometry in which they are false.²⁷ Developing such a model is the primary purpose of this chapter. Its existence explains why Saccheri and Lambert ultimate goal was unrealized: the parallel postulate (Playfair) is indeed independent of Euclid's (Hilbert's) other axioms.

The Saccheri–Legendre Theorem

We turn now to Saccheri's consideration of triangles, suitably modernized. We work in absolute geometry, starting with an extension of Euclid's proof of the exterior angle theorem.

Suppose $\triangle ABC$ has angle-sum $\Sigma_{\triangle}$, let $\alpha_0 = m\angle ABC$ and construct M and E following Euclid. Observe:



1. $\angle ACB + \angle CAB \cong \angle ACB + \angle ACE < 180^\circ$ is the exterior angle theorem. More generally, the exterior angle theorem guarantees that the *sum of any two angles in a triangle is strictly less than 180°* .
2. $\triangle ABC$ and $\triangle EBC$ have the *same angle-sum*—just look at the picture!

$$\Sigma_{\triangle} = \text{red} + \text{green} + \text{blue} + \text{pink}$$

Remember that we do not know whether $\Sigma_{\triangle} = 180^\circ$!

3. $\triangle EBC$ has at least one angle ($\angle EBC$ or $\angle BEC$) measuring at most $\frac{1}{2}\alpha_0 = \frac{1}{2}m\angle ABC$.

Iterate this construction: if $\angle EBC$ is smaller, start by bisecting $\overline{CE}$; otherwise bisect $\overline{BC}$

The result is an infinite sequence of triangles $\triangle_1 = \triangle EBC$, $\triangle_2$, $\triangle_3$, ... with two properties:

- (a) All triangles in the sequence have *same angle-sum* $\Sigma_{\triangle} = \Sigma_{\triangle_1} = \Sigma_{\triangle_2} = \dots$.
- (b) The n^{th} triangle $\triangle_n$ has at least one angle measuring $\alpha_n \leq \frac{1}{2^n}\alpha_0$.

Now suppose, for contradiction, that $\Sigma_{\triangle} = 180^\circ + \epsilon$ is *strictly greater* than 180° . Since $\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$, we may choose n large enough to guarantee $\alpha_n < \epsilon$:

$$n > \log_2 \frac{\alpha_0}{\epsilon} \implies \alpha_n < \epsilon$$

But then the sum of the *other two* angles in $\triangle_n$ would be *greater than* 180° , contradicting the exterior angle theorem (observation 1)! We have proved a famous result.

Theorem 4.2 (Saccheri–Legendre). *Triangles in absolute geometry have angle-sum $\Sigma_{\triangle} \leq 180^\circ$.*

Saccheri's failed hope was to prove *equality* without invoking the parallel postulate.

²⁷The model must be of *absolute* geometry. For instance, the angles in a spherical triangle sum to more than 180° , but we can't guarantee that the reason for this is *only* the lack of parallels, since spherical geometry fails two of Hilbert's other axioms.

Saccheri and Lambert Quadrilaterals

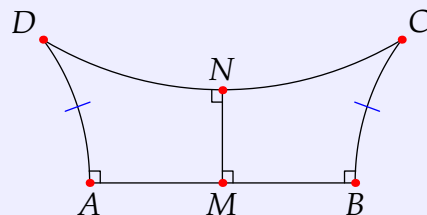
Two families of quadrilaterals in absolute geometry are named in honor of these pioneers.

Definition 4.3. In absolute geometry, a *Lambert quadrilateral* $AMND$ has three right-angles.

A *Saccheri quadrilateral* $ABCD$ satisfies

$$\overline{AD} \cong \overline{BC} \quad \text{and} \quad \angle DAB = \angle CBA = 90^\circ$$

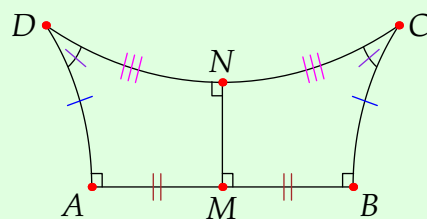
$\overline{AB}$ is the *base* and $\overline{CD}$ the *summit*. The interior angles at C and D are the *summit angles*.



We draw these with curved sides to indicate that the summit angles need not be right-angles, though we haven't yet exhibited a model which shows they could be anything else. Regardless of how they are drawn, $\overline{AD}$, $\overline{BC}$ and $\overline{CD}$ are all *segments*!

Lemma 4.4. *Saccheri and Lambert quadrilaterals exhibit basic symmetry as pictured.*

1. If the base and summit of a Saccheri quadrilateral are bisected, we obtain two congruent Lambert quadrilaterals.
2. The summit angles of a Saccheri quadrilateral are congruent.
3. In **Euclidean geometry**, Saccheri and Lambert quadrilaterals are **rectangles**: the summit angles are also right-angles, and the summit and base are congruent.



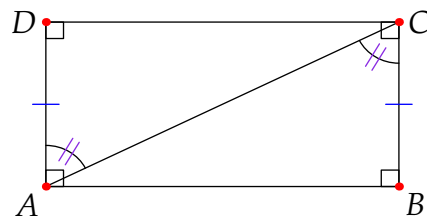
We could interpret part 3 as saying that Saccheri and Lambert quadrilaterals are as close as we can get to rectangles in absolute geometry. The Lemma is mostly a recapitulation of Euclid's Book I construction of a square (Thm. I. 46, Exercise 2.1.2). We leave parts 1 and 2 as exercises: remember that parts 1 and 2 must be proved without reference to Playfair's Axiom!.

Proof of part 3. Given a Lambert quadrilateral, a second congruent copy forms a Saccheri quadrilateral: if this last is a rectangle, so is the former. It suffices therefore to prove the result for Saccheri quadrilaterals.

Suppose $ABCD$ is a Saccheri quadrilateral with base $\overline{AB}$. Following the exterior angle theorem, since $\overleftrightarrow{AB}$ is a crossing line making congruent right-angles we conclude that $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$.

The extended summit $\overleftrightarrow{CD}$ also crosses the same parallels. By the parallel postulate, the summit angles sum to a straight-edge. Since these are congruent (part 2), both are right-angles.

Finally, join $\overline{AC}$. Again by the parallel postulate, $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$ says the alternate angles are congruent ($\angle DAC \cong \angle BCA$). By SAS the triangles are congruent ($\triangle DAC \cong \triangle BCA$); in particular so also are the summit and base ($\overline{AB} \cong \overline{CD}$) of the original quadrilateral.



We now show that drawing a Saccheri quadrilateral with *non-obtuse* summit angles is justified.

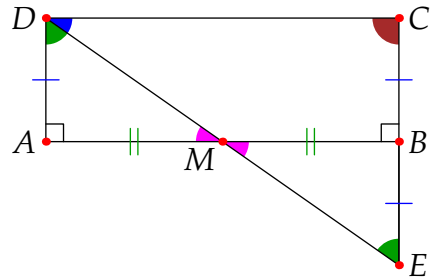
Theorem 4.5. *In absolute geometry, the summit angles of a Saccheri quadrilateral measure $\leq 90^\circ$.*

Proof. Suppose $ABCD$ is a Saccheri quadrilateral with base $\overline{AB}$.

Extend $\overline{CB}$ to E such that $\overline{BE} \cong \overline{DA}$.

Let M be the midpoint of $\overline{AB}$ and join to D and E .

SAS implies $\angle DAM \cong \angle EBM$. In particular, the **vertical angles** at M are congruent, whence M lies on $\overline{DE}$.



By the Saccheri–Legendre Theorem, the (congruent) **summit angles** at C and D sum to

$$\begin{aligned}\angle ADC + \angle DCB &= \angle ADM + \angle EDC + \angle DCE \\ &= \angle CED + \angle EDC + \angle DCE \\ &\leq 180^\circ\end{aligned}$$

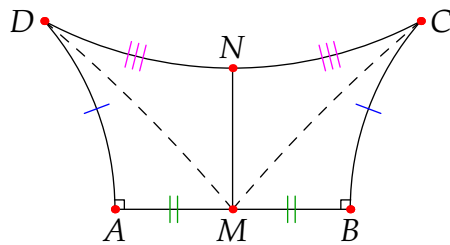
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The next step is to see if there exists a model of absolute geometry in which the summit angles are *strictly less* than a right-angle...

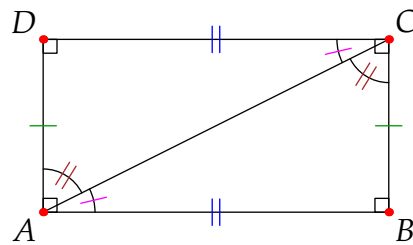
Exercises 4.1. *Key concepts: Absolute Geometry, $\Sigma_\Delta \leq 180^\circ$, Saccheri & Lambert quadrilaterals*

Work in absolute geometry; you cannot use Playfair's Axiom or the parallel postulate!

1. Use the first picture below to prove parts 1 and 2 of Lemma 4.4.
2. Use the first picture to give an alternative proof of Theorem 4.5.



Exercises 1 & 2



Exercise 3

3. Suppose $ABCD$ has four right-angles. Use the Saccheri–Legendre Theorem to prove that $\overline{AC}$ splits $ABCD$ into two congruent triangles; conclude that $ABCD$ is a rectangle.
(This is easy in Euclidean geometry, but shows that we needn't have invoked the parallel postulate for a second time in Lemma 4.4, part 3.)
4. Suppose a pair of Saccheri quadrilaterals have congruent bases and perpendicular sides. Prove that the quadrilaterals are congruent.
5. (Hard) Suppose two Saccheri quadrilaterals have congruent summits and congruent perpendicular sides. Prove that the quadrilaterals are congruent.
(Hint: It is not necessary, but you might find it easier to think this out using Lambert quadrilaterals)

4.2 Models of Hyperbolic Geometry

In the 1820-30s, János Bolyai, Carl Friedrich Gauss and Nikolai Lobachevsky independently took the next step.²⁸ Rather than attempting to establish the parallel postulate as a theorem within Euclidean geometry, they defined a new geometry based on an alternative axiom of parallels.

Axiom 4.6 (Bolyai–Lobachevsky/Hyperbolic Postulate). Given a line ℓ and a point $P \notin \ell$, there exist *at least two* parallel lines to ℓ through P .

Hyperbolic Geometry is the resulting axiomatic system: Hilbert with Playfair's axiom replaced by the hyperbolic postulate. Consistency was proved in the late 1800s by Beltrami, Klein and Poincaré, each of whom created models by defining *point*, *line*, etc., in novel ways.

The Poincaré Disk Model

One of the simplest models is named for Poincaré, though it was first proposed by Beltrami.²⁹

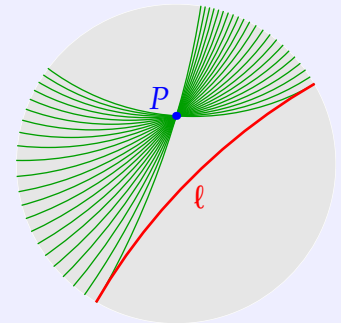
Definition 4.7. The *Poincaré disk* is the interior of the unit circle

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\} \quad \text{or} \quad \{z \in \mathbb{C} : |z| < 1\}$$

A *hyperbolic line* is either a diameter or a circular arc meeting the unit circle at right-angles.

The picture shows a **hyperbolic line** ℓ and a **point** P : also drawn are several **parallel hyperbolic lines** to ℓ passing through P .

Points on the boundary circle are termed *omega-points*: these are *not* in the Poincaré disk and are essentially 'points at infinity.'



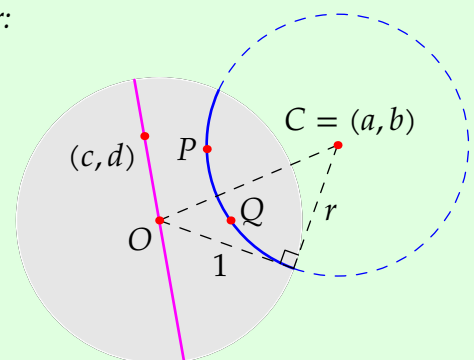
By the incidence axioms, there exists a unique hyperbolic line joining any two points in the Poincaré disk. Such may straightforwardly be described using equations in analytic geometry.

Lemma 4.8. A hyperbolic line in the Poincaré disk model is either:

1. A **diameter** with Euclidean equation $dx = cy$ (passes through a point $(c, d) \neq (0, 0)$).
2. An **arc** of a Euclidean circle with equation

$$x^2 + y^2 - 2ax - 2by + 1 = 0 \quad \text{where} \quad a^2 + b^2 > 1$$

Euclidean center $C = (a, b)$ and radius $r = \sqrt{a^2 + b^2 - 1}$.



²⁸Bolyai is the source of the term 'absolute geometry.'

²⁹The key results of hyperbolic geometry can be discussed synthetically without reference to a model, though such an approach would be ahistorical, and masochistic for a first exposure!

Example 4.9. We compute the equation of the hyperbolic line through the points $P = (\frac{1}{3}, \frac{1}{2})$ and $Q = (\frac{1}{2}, 0)$ in the Poincaré disk: this is the picture shown in Lemma 4.8.

This requires only high-school algebra! Substitute the given points into the equation for a hyperbolic line ($x^2 + y^2 - 2ax - 2by + 1 = 0$) to obtain equations for the coefficients a, b :

$$\begin{cases} \frac{1}{9} + \frac{1}{4} - \frac{2}{3}a - b + 1 = 0 \\ \frac{1}{4} - a + 1 = 0 \end{cases} \implies (a, b) = \left(\frac{5}{4}, \frac{19}{36}\right)$$

The required hyperbolic line $\overleftrightarrow{PQ}$ therefore has equation

$$x^2 + y^2 - \frac{5}{2}x - \frac{19}{18}y + 1 = 0 \quad \text{or} \quad \left(x - \frac{5}{4}\right)^2 + \left(y - \frac{19}{36}\right)^2 = \frac{545}{648}$$

This is an arc of a Euclidean circle centered at $C = (\frac{5}{4}, \frac{19}{36})$ with radius $r = \sqrt{\frac{545}{648}}$.

Distance and Angle in the Poincaré Disk The undefined terms *point*, *line*, *on* and *between* now make sense. To complete the model, we need to define *congruence* of segments and angles.

Definition 4.10. The *hyperbolic distance* between points P, Q in the Poincaré disk is³⁰

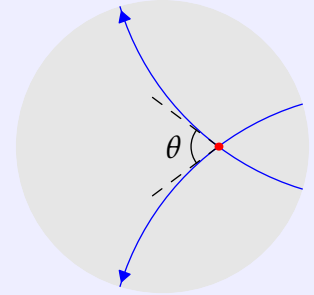
$$d(P, Q) := \cosh^{-1} \left(1 + \frac{2|PQ|^2}{(1 - |P|^2)(1 - |Q|^2)} \right)$$

where $|PQ|$, $|P| = |OP|$ and $|Q| = |OQ|$ are Euclidean distances. Hyperbolic segments are *congruent* if they have the same length

$$\overline{AB} \cong \overline{PQ} \iff d(A, B) = d(P, Q)$$

The *angle* between hyperbolic rays is that between their Euclidean tangent rays. Angles are *congruent* if they are so in a Euclidean sense.

The *hyperbolic circle* with center P and (hyperbolic) radius ρ is the set of points X for which $d(X, P) = \rho$.



It is worth observing both the symmetry of the distance function ($d(P, Q) = d(Q, P)$) and how it simplifies when measuring from the origin: if $Q = O$, then

$$d(O, P) = \cosh^{-1} \frac{1 + |P|^2}{1 - |P|^2} = \ln \frac{1 + |P|}{1 - |P|}$$

³⁰It should seem reasonable for hyperbolic functions to play some role in hyperbolic geometry! For reference:

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh^2 x - \sinh^2 x = 1, \quad \cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$$

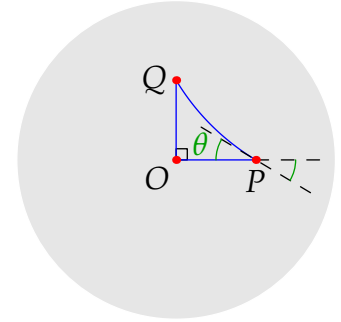
Example 4.11. We calculate the side-lengths and angles in the hyperbolic isosceles right-triangle with vertices $O = (0,0)$, $P = (\frac{1}{2}, 0)$ and $Q = (0, \frac{1}{2})$.

To compute the hyperbolic distances, we first need the three *Euclidean* versions:

$$|P| = \frac{1}{2} = |Q|, \quad |PQ|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$d(O, P) = d(O, Q) = \ln \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} = \ln 3 = \cosh^{-1} \frac{5}{3} \approx 1.10$$

$$d(P, Q) = \cosh^{-1} \left(1 + \frac{2 \cdot \frac{1}{2}}{(1 - \frac{1}{4})^2} \right) = \cosh^{-1} \frac{25}{9} \approx 1.68$$



To find the interior angle θ , implicitly differentiate the equation for the hyperbolic line $\overleftrightarrow{PQ}$:

$$x^2 + y^2 - \frac{5}{2}x - \frac{5}{2}y + 1 = 0 \Rightarrow \frac{dy}{dx} \Big|_P = \frac{4x - 5}{5 - 4y} \Big|_P = -\frac{3}{5} \Rightarrow \theta = \tan^{-1} \frac{3}{5} \approx 31^\circ$$

By symmetry, we have the same angle at Q . With a right-angle at O , we conclude that the angle-sum is $\Sigma_\Delta \approx 152^\circ$! To satisfy curiosity, we compare data for $\triangle OPQ$ and the *Euclidean* triangle with the same vertices

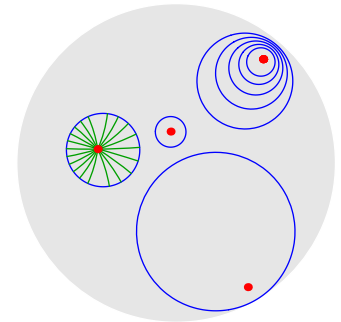
Property	Hyperbolic Triangle	Euclidean Triangle
Edge lengths	1.10 : 1.10 : 1.68	0.5 : 0.5 : 0.71
Relative edge ratios	1 : 1 : 1.53	1 : 1 : 1.41
Angles	31°, 31°, 90°	45°, 45°, 90°

The hyperbolic triangle has longer sides and a *relatively* longer hypotenuse. Moreover, its side lengths do *not* satisfy the Pythagorean relation³¹ $a^2 + b^2 = c^2$.

The Poincaré disk uses concepts from *Euclidean* (analytic) geometry to define objects and compute in *hyperbolic* geometry. It may seem laborious to keep stressing the distinction, but it is essential not to confuse the corresponding versions of distance, line, circle, etc. While the meaning of *angle* is unchanged, everything else is at least slightly, if not significantly, different.

For instance, the picture shows several **hyperbolic circles** and their **centers**; one has several of its **radii** drawn. Observe how the centers are closer (in a Euclidean sense) to the boundary circle (omega-points) than one might expect; this is since hyperbolic distance measure is greater the further one is from the origin.

In fact (Exercise 6) hyperbolic circles in the Poincaré disk model are also Euclidean circles! Their hyperbolic **radii** moreover meet the circles at right-angles, as we'd expect.



³¹Though the relationship $\cosh a \cosh b = \cosh c$ should be obvious. We'll return to this observation later...

A Little Bookkeeping The next result is an exercise; it says that the distance function behaves roughly as we expect, increasing smoothly as one moves along a hyperbolic line.

Lemma 4.12. Fix P and a hyperbolic line ℓ through P . Then the distance function $Q \mapsto d(P, Q)$ maps the points of ℓ on one side of P differentiably and bijectively onto the interval $(0, \infty)$.

This should also convince us that hyperbolic circles look roughly as we expect. Of greater importance is the following.

Theorem 4.13. The Poincaré disk is a model of hyperbolic geometry.

Sketch Proof. A rigorous proof would require us to check the hyperbolic postulate and all Hilbert's axioms except Playfair. Instead we verify Euclid's postulates 1–4 and the hyperbolic postulate 5.

1. Lemma 4.8 says we can join any given points in the Poincaré disk by a unique segment.
2. A hyperbolic segment joins two points *inside* the (open) Poincaré disk. The distance formula increases (Lemma 4.12) unboundedly as P moves towards the boundary circle, so we can always make a hyperbolic segment longer.
3. Hyperbolic circles are as in Definition 4.10.
4. All right-angles are equal since the notion of angle is unchanged from Euclidean geometry.
5. The first picture on page 69 shows multiple parallels!

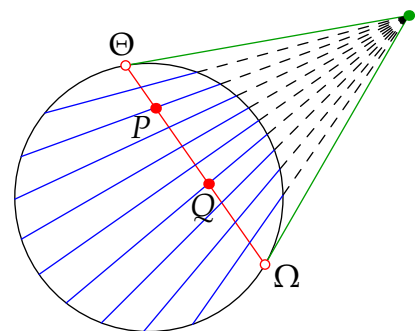
Other Models of Hyperbolic Space (non-examinable)

There are several other models of hyperbolic space. Here are three of the most common.

Klein Disk Model This is similar to the Poincaré disk. Instead of Euclidean arcs, hyperbolic lines are chords of the unit circle ('Euclidean' line-segments) without their end-points, and the distance function is different: if Ω, Θ are the omega-points of the chord $\overleftrightarrow{PQ}$, then

$$d_K(P, Q) = \frac{1}{2} \left| \ln \frac{|P\Theta||Q\Omega|}{|P\Omega||Q\Theta|} \right|$$

Lines might look "straight", but the cost is that the notion of *angle* is different.³² Given a **hyperbolic line** (as pictured), find the **tangents** at its omega-points; any **chord** whose extension passes through the **intersection** of these tangents is perpendicular to the **original line**. Measuring other angles is difficult!

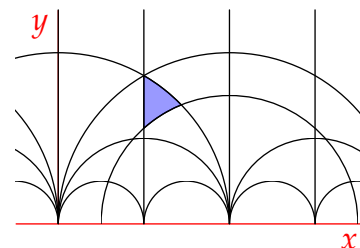


³²For differential geometers, Gauss' famous *theorem egregium* says that this problem is unavoidable. The intrinsic (negative) curvature of hyperbolic geometry says that there is no model in which both lines and angles have the same meanings as in Euclidean geometry.

Poincaré Half-plane Model Widely used in complex analysis, the points comprise the upper half-plane ($y > 0$) in $\mathbb{R}^2$, while hyperbolic lines are verticals or semicircles centered on the x -axis

$$x = \text{constant} \quad \text{or} \quad (x - a)^2 + y^2 = r^2$$

and angles are the same as in Euclidean space. The expression for hyperbolic distance remains horrific! The picture shows several hyperbolic lines and a **hyperbolic triangle**.

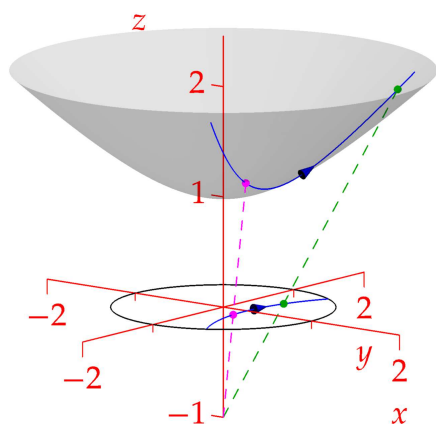


Hyperboloid Model Points comprise the upper sheet ($z \geq 1$) of the hyperboloid $x^2 + y^2 = z^2 - 1$. A **hyperbolic line** is the intersection of the hyperboloid with a plane through the origin. Isometries (congruence) can be described using matrix-multiplication and hyperbolic distance is relatively easy: given $P = (x, y, z)$ and $Q = (a, b, c)$,

$$d(P, Q) = \cosh^{-1}(cz - ax - by)$$

An obvious drawback is the need to work in three-dimensions. Moreover, angles are awkward.

The relationship to the Poincaré disk is via projection. Place the disk in the xy -plane centered at the origin and draw a **line** through the disk and the point $(0, 0, -1)$. The intersection of this line with the hyperboloid gives the correspondence.



Exercises 4.2. Key concepts: Hyperbolic Postulate, Poincaré disk, Hyperbolic lines and distance

Answer all questions within the Poincaré disk model.

- Find the equation of the hyperbolic line through $P = (\frac{1}{4}, 0)$ and $Q = (0, \frac{1}{2})$.
 - Find the side lengths of the hyperbolic triangle $\triangle OPQ$ where $O = (0, 0)$ is the origin.
 - $\triangle OPQ$ is right-angled at O . If o, p, q represent the hyperbolic lengths of the sides opposite O, P, Q respectively, check that $p^2 + q^2 \neq o^2$ is *false*. Now compute $\cosh p \cosh q$: what do you observe?
- Find the omega points for the hyperbolic line with equation $x^2 + y^2 - 4x + 10y + 1 = 0$.
- Let $P = (\frac{1}{2}, \sqrt{\frac{5}{12}})$ and $Q = (\frac{1}{2}, -\sqrt{\frac{5}{12}})$. Together with the origin O these form a triangle.
 - Compute the hyperbolic distances $d(O, P)$, $d(O, Q)$ and $d(P, Q)$.
 - Compute the interior angle at O ($\angle POQ$).
 - Show that the hyperbolic line $\ell = \overline{PQ}$ has equation $x^2 - \frac{10}{3}x + y^2 + 1 = 0$.
 - Calculate $\frac{dy}{dx}$ to show that a tangent vector to ℓ at P is $\sqrt{15}\mathbf{i} + 7\mathbf{j}$. Hence compute the interior angle at P ($\angle OPQ$). What is the angle-sum in this triangle?

4. We generalize Example 4.11. Suppose $0 < c < 1$ and consider the isosceles right-triangle with vertices $O = (0, 0)$, $P = (c, 0)$ and $Q = (0, c)$ in the Poincaré disk.

- (a) Compute the hyperbolic side lengths of $\triangle OPQ$ as functions of c .
- (b) Find the equation of the hyperbolic line joining $P = (c, 0)$ and $Q = (0, c)$.
- (c) Use implicit differentiation to prove that the interior angles at P and Q measure $\tan^{-1} \frac{1-c^2}{1+c^2}$. What happens as $c \rightarrow 0^+$ and as $c \rightarrow 1^-$?

5. Let $0 < r < 1$. Consider the hyperbolic triangle with vertices

$$P = (r, 0), \quad Q = \left(-\frac{r}{2}, \frac{\sqrt{3}r}{2}\right), \quad R = \left(-\frac{r}{2}, -\frac{\sqrt{3}r}{2}\right)$$

- (a) Find the side-lengths and thus confirm that $\triangle PQR$ is equilateral.
 - (b) Compute the interior angles of $\triangle PQR$ as functions of r .
 - (c) What happens to the triangle (side-lengths, angles) as $r \rightarrow 0^+$? As $r \rightarrow 1^-$?
6. (a) Prove that the hyperbolic circle with hyperbolic radius $\rho = \ln 3$ and center $C = (\frac{1}{2}, 0)$ has *Euclidean* equation

$$\left(x - \frac{2}{5}\right)^2 + y^2 = \frac{4}{25}$$

- (b) Prove that every hyperbolic circle in the Poincaré disk model is also a Euclidean circle. Conversely, what restrictions on the constants p, q, r guarantee that a Euclidean circle with equation $(x - p)^2 + (y - a)^2 = r^2$ describes a hyperbolic circle?

7. We sketch a proof of Lemma 4.12.

- (a) Prove that $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$ is strictly increasing on the interval $(1, \infty)$.
- (b) By part (a), it is enough to show that $\frac{|PQ|^2}{1-|Q|^2}$ increases as Q moves away from P along a hyperbolic line. Appealing to symmetry, let $P = (0, c)$ lie on the hyperbolic line with equation $x^2 + y^2 - 2by + 1 = 0$. If $Q = (x, y)$ also lies on the same line, prove that

$$\frac{|PQ|^2}{1-|Q|^2} = \frac{(b-c)y + bc - 1}{1-by}$$

and show that this is an increasing function of y when $c < y < \frac{1}{b}$.

4.3 Parallels, Perpendiculars & Angle-Sums

From now on, all examples will be illustrated within the Poincaré disk, though the main results hold in any model. Recall (page 65) that we may use anything from absolute geometry; as a sanity check, think through how the picture illustrates the following familiar result.

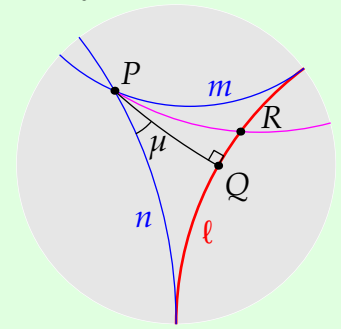
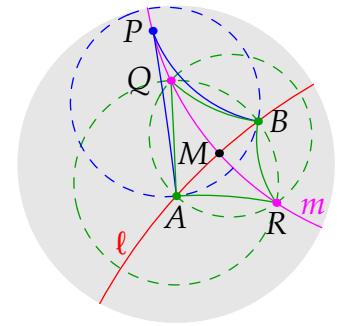
Lemma 4.14. *Through a point P not on a line ℓ there exists a unique perpendicular to ℓ .*

We now consider a major departure from Euclidean geometry.

Theorem 4.15 (Fundamental Theorem of Parallels in Hyperbolic Geometry).

Given $P \notin \ell$, drop the perpendicular $\overline{PQ}$ to ℓ . Then there exist precisely two **limiting**, or **asymptotic**, **parallels** m, n to ℓ through P with the following properties:

1. A ray based at P intersects ℓ if and only if it lies between m and n in the same fashion as $\overrightarrow{PQ}$.
2. The limiting parallels make congruent acute angles μ with $\overrightarrow{PQ}$, the **angle of parallelism** at P relative to ℓ .



Every other (non-asymptotic) parallel is called an *ultraparallel*. More generally, parallel lines ℓ, m are *asymptotic* if they ‘meet’ at an omega-point.

The proof depends crucially on ideas from analysis, particularly continuity & suprema. As you read through, consider how everything *except* the last line is valid in Euclidean geometry!

Proof. Points $R \in \ell$ are in continuous bijective correspondence with the real numbers (Lemma 4.12). We therefore have a *continuous, increasing* function

$$f : \mathbb{R} \rightarrow (-90^\circ, 90^\circ) \quad \text{where} \quad f(r) = \angle QPR \quad (\pm 90^\circ \notin \text{range } f \text{ by Saccheri–Legendre})$$

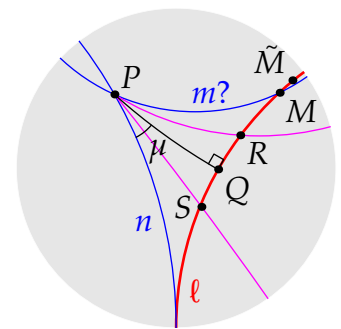
Since $\text{dom } f = \mathbb{R}$ is an interval, the intermediate value theorem says that $\text{range } f$ is a *subinterval* $I \subseteq (-90^\circ, 90^\circ)$.

Given $R \in \ell$, transfer $\overline{QR}$ to the other side of Q to obtain $S \in \ell$. By SAS, $\angle QPS = -\angle QPR$ whence I is *symmetric*: $\theta \in I \iff -\theta \in I$.

Let $\mu := \sup I \in (0^\circ, 90^\circ]$ be the least upper bound; by symmetry, $\inf I = -\mu$. Let m and n be the lines making angles $\pm\mu$ respectively. Certainly any ray between m, n intersects ℓ . To finish part 1, it remains only to see that m, n are parallel to ℓ (that is, $I = (-\mu, \mu)$).

For contradiction, suppose m intersects ℓ at M . Transfer $\overline{QM}$ to the other side of M to obtain $\tilde{M}$. Since f is increasing, $\angle QP\tilde{M} > \mu = \angle QPM$ contradicts $\mu = \sup I$.

Finally $m = n \iff \mu = 90^\circ$. In such a case there would exist only one parallel to ℓ through P , contradicting the hyperbolic postulate. ■



The picture suggests a bijective relationship between μ and the perpendicular distance. Here it is; a simplified argument is in Exercise 4.3.6, and the full result to the next section.

Corollary 4.16. *The perpendicular distance $\delta = d(P, Q)$ and the angle of parallelism are related via*

$$\cosh \delta = \csc \mu \quad \text{or equivalently} \quad \tan \frac{\mu}{2} = e^{-\delta}$$

Examples 4.17. We compute the limiting parallels and angles of parallelism in two cases.

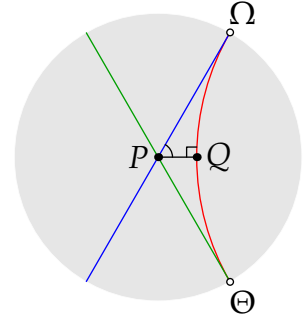
1. Let ℓ be the hyperbolic line with equation $x^2 + y^2 - 4x + 1 = 0$. Intersect with $x^2 + y^2 = 1$ to find the omega-points $\Omega = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ and $\Theta = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$.

By symmetry, the perpendicular $\overline{PQ}$ from the origin P has equation $y = 0$, resulting in $Q = (2 - \sqrt{3}, 0)$.

The limiting parallels through P have equations $y = \pm\sqrt{3}x$, from which the angle of parallelism is $\mu = \tan^{-1} \sqrt{3} = 60^\circ$.

In accordance with Corollary 4.16, we easily verify that

$$\delta = d(P, Q) = \ln \frac{1 + (2 - \sqrt{3})}{1 - (2 - \sqrt{3})} = \ln \sqrt{3} \Leftrightarrow e^{-\delta} = \frac{1}{\sqrt{3}} = \tan \frac{\mu}{2}$$



2. Let $P = \left(-\frac{3}{10}, \frac{4}{10}\right)$ relative to the **hyperbolic line** with equation $x^2 + y^2 + 2x + 4y + 1 = 0$. First find the omega-points by intersecting with $x^2 + y^2 = 1$:

$$\Omega = (-1, 0), \quad \Theta = \left(\frac{3}{5}, -\frac{4}{5}\right)$$

Plainly $\overline{P\Theta}$ is the diameter $y = -\frac{4}{3}x$ with slope $-\frac{4}{3}$.

For $\overline{P\Omega}$, substitute these points into the usual expression for a hyperbolic line (Lemma 4.8); implicitly differentiate for the **slope**:

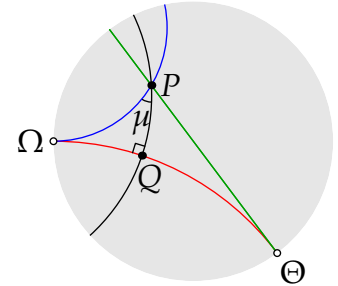
$$x^2 + y^2 + 2x - \frac{13}{8}y + 1 = 0 \Rightarrow \left. \frac{dy}{dx} \right|_P = \frac{16(1+x)}{13-16y} \Big|_P = \frac{16 \cdot \frac{7}{10}}{13 - \frac{64}{10}} = \frac{56}{33}$$

The angle of parallelism is *half* that between the tangent vectors $\begin{pmatrix} -33 \\ -56 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$:

$$\mu = \frac{1}{2} \cos^{-1} \frac{\begin{pmatrix} -33 \\ -56 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \end{pmatrix}}{\left| \begin{pmatrix} -33 \\ -56 \end{pmatrix} \right| \left| \begin{pmatrix} 3 \\ -4 \end{pmatrix} \right|} = \frac{1}{2} \cos^{-1} \frac{5}{13} \approx 33.7^\circ$$

Corollary 4.16 can now be used to find the perpendicular distance $d(P, Q) = \ln \frac{3+\sqrt{13}}{2}$.

If you want a major challenge, try to verify that $Q = \left(\frac{93(-29+2\sqrt{117})}{1865}, \frac{26(-29+2\sqrt{117})}{1865} \right)$.

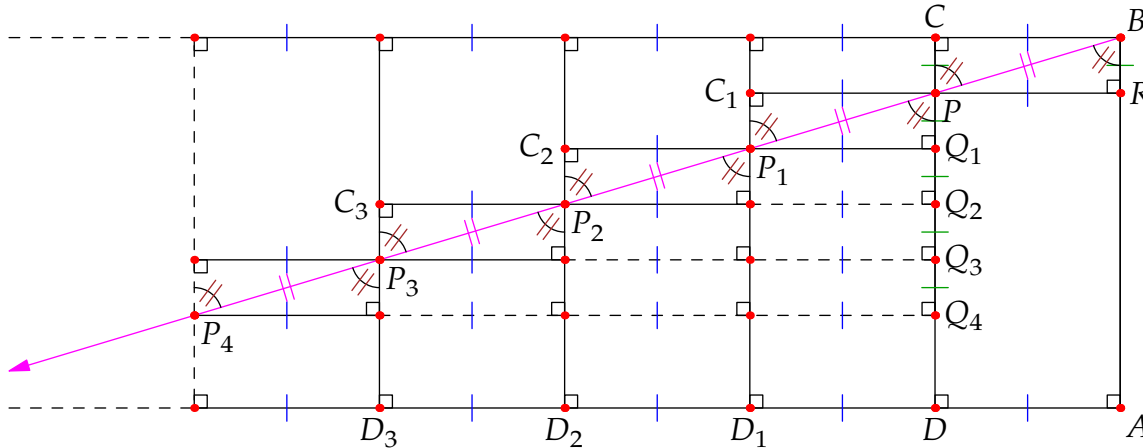


Angles in Triangles, Rectangles and the AAA Congruence

We at last come to the triple punchline. Everything follows from one big picture.

Theorem 4.18. Suppose $\square ABCD$ is a rectangle in absolute geometry. Then the angle of parallelism of B with respect to $\ell := \overrightarrow{AD}$ is 90° . Otherwise said, $\overrightarrow{BC}$ is the unique parallel to $\overrightarrow{AD}$ through B .

Proof. Reflect $\square ABCD$ across $\overline{CD}$ and repeat to obtain an infinite family of congruent rectangles (Exercise 4.1.4). Let $P \in \overline{CD}$ and drop perpendiculars to $R \in \overline{AB}$ and C_1 as shown.



We claim $\square PRBC$ is a rectangle: if not, then one of $\square ARPD$ or $\square PRBC$ would have angle-sum exceeding 360° and contradicting Saccheri–Legendre. Similarly $\square DPC_1D_1$ is a rectangle.

By Exercise 4.1.3, $\overrightarrow{BP}$ splits $\square PRBC$ into a pair of congruent triangles. In particular, $\overrightarrow{BP}$ crosses $\overline{CD}$ at the **same angle** it leaves B .

Iterate to obtain the picture: the small rectangles are congruent (essentially Exercise 4.1.4 again), whence $P, Q_1, Q_2, \dots$ is an *equidistant* sequence along $\overline{CD}$.

Since $\overline{CD}$ is *finite*, the Archimedean property (Exercise 2.4.8) says that this sequence eventually passes D : some Q_n lies on the opposite side of ℓ to B , whence $P_n \in \overrightarrow{BP}$ does also. We conclude that $\overrightarrow{BP}$ intersects ℓ .

Since $P \in \overline{CD}$ was generic, the angle of parallelism of B with respect to ℓ is 90° . ■

Corollary 4.19. Within hyperbolic geometry:

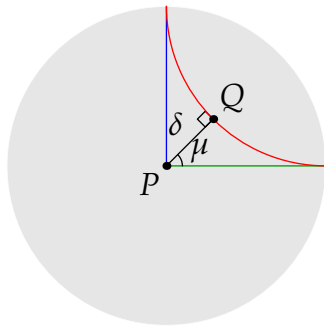
1. There are **no rectangles** (quadrilaterals with four right-angles). In particular, the summit angles of a Saccheri quadrilateral are **acute** (compare Theorem 4.5).
2. The angles in a triangle sum to **strictly less than** 180° .
3. (AAA congruence) If the angles of $\triangle ABC$ and $\triangle DEF$ are congruent in pairs, then the triangles are **congruent** ($\triangle ABC \cong \triangle DEF$).

Part 1 is simply the Theorem: a unique parallel contradicts the hyperbolic postulate. We leave the rest to Exercises 7 and 8. Notice that AAA is a *congruence* theorem in hyperbolic geometry,

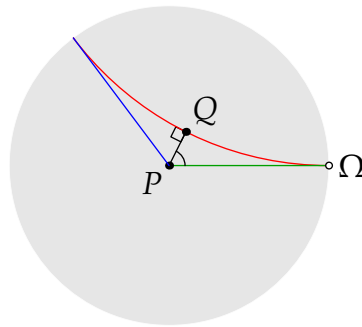
not a *similarity* theorem (compare Theorem 2.43). Finally, revisit the observations on page 65; Euclid's arguments making use of the parallel postulate genuinely *require* it!

Exercises 4.3. *Key concepts: Limiting parallels, Angle of parallelism, $\cosh \delta = \csc \mu$, No rectangles, $\Sigma_{\Delta} < 180^\circ$, AAA congruence*

1. Use Corollary 4.19 to prove the following within hyperbolic geometry.
 - (a) Two hyperbolic lines cannot have more than one common perpendicular.
 - (b) Saccheri quadrilaterals with congruent summits and summit angles are congruent.
2. Suppose P lies a perpendicular distance $\delta = d(P, Q) = \ln \sqrt{3} = \frac{1}{2} \ln 3$ from a hyperbolic line ℓ , and that the ray $\overrightarrow{PR}$ makes angle 45° with the perpendicular $\overrightarrow{PQ}$. Determine whether $\overrightarrow{PQ}$ intersects ℓ , is a limiting parallel, or an ultraparallel.
3. Suppose ℓ intersects m at a right-angle and that m, n are parallel.
 - (a) In *Euclidean geometry*, prove that ℓ intersects n at a right-angle.
 - (b) What are the possible arrangements in *hyperbolic geometry*? Draw some pictures.
4. Let ℓ be the hyperbolic line with equation $(x - 1)^2 + (y - 1)^2 = 1$, let P be the origin and drop the perpendicular $\overrightarrow{PQ}$ to $Q \in \ell$.
 - (a) Explain why the angle of parallelism of P with respect to ℓ is 45° and why Q has co-ordinates $(1 - \frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}})$.
 - (b) Use the hyperbolic distance formula (Definition 4.10) to compute the perpendicular distance from ℓ to the origin. Hence verify the relation $\cosh \delta = \csc \mu$.



Question 4



Question 5

5. Similarly to Question 4, verify the relation $\cosh \delta = \csc \mu$ for the origin P relative to the hyperbolic line ℓ with equation $x^2 + y^2 - 2x - 4y + 1 = 0$.
 (Hints: Where is the 'center' of ℓ viewed as a Euclidean circle? As in Question 4, $\Omega = (1, 0)$ is an omega-point for ℓ .)

6. We generalize Example 4.17.1. Suppose $P = (0, 0)$ is the origin and let $Q = (r, 0)$ where $0 < r < 1$. Let ℓ be the hyperbolic line passing through Q at right-angles to $\overline{PQ}$.

(a) Find the equation of ℓ and prove that its limiting parallels through P have equations

$$\pm 2ry = (1 - r^2)x$$

(b) Let μ be the angle of parallelism of P relative to ℓ and $\delta = d(P, Q)$ the hyperbolic distance. Prove the special case of Corollary 4.16, that $\cosh \delta = \csc \mu$.

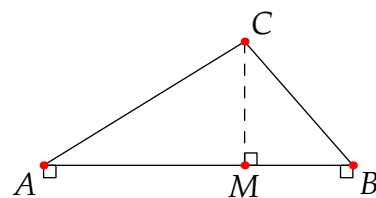
(Hint: $\csc^2 \mu = 1 + \cot^2 \mu = 1 + \frac{1}{\tan^2 \mu} = \dots$)

(c) By differentiating $\cosh \delta = \csc \mu$, verify the claim that δ and μ are bijectively related.

7. We work in absolute geometry. Suppose $\triangle ABC$ has longest side $\overline{AB}$ (the other sides are no larger) and drop the perpendicular from C to $M \in \overline{AB}$.

(a) Prove that M is interior to $\overline{AB}$ by showing that the other possibilities are contradictions.

(b) Suppose a triangle exists with angle-sum 180° . Show that there exists such a *right-triangle*, and therefore a rectangle. Hence conclude part 2 of Corollary 4.19.



(c) Explain why part (a) is needed to prove (b): what might happen if $\overline{AB}$ isn't the longest side?

8. We prove the AAA congruence theorem in hyperbolic geometry (Corollary 4.19, part 3).

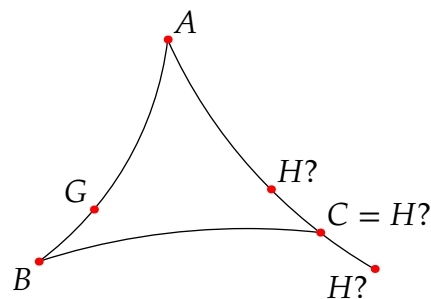
Suppose, for contradiction, that *non-congruent* triangles $\triangle ABC$ and $\triangle DEF$ have angles congruent in pairs ($\angle A \cong \angle D$, etc.). Without loss of generality, assume $\overline{DE} < \overline{AB}$. By segment transfer, there exist unique points:

- G interior to $\overline{AB}$ such that $\overline{DE} \cong \overline{AG}$.
- $H \in \overrightarrow{AC}$ such that $\overline{DF} \cong \overline{AH}$.

(a) Explain why $\triangle DEF \cong \triangle AGH$.

(b) There are three possible locations for H .

- i. H is interior to $\overline{AC}$.
- ii. $H = C$.
- iii. C lies between A and H .



By connecting $\overline{GH}$, in each case explain why we have a contradiction.

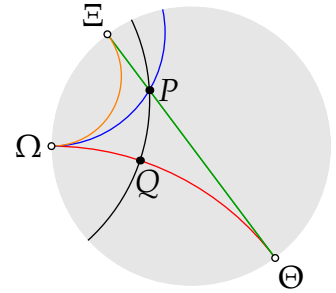
4.4 Omega-triangles

Recall that limiting parallels (Theorem 4.15) ‘meet’ at an omega-point.

Definition 4.20. An *omega-triangle* or *ideal-triangle* is a ‘triangle’ where at least two sides are limiting parallels. Alternatively (in the Poincaré disk model), one or more of the vertices is an omega-point.

The three types of omega-triangle depend on how many omega-points they have. In the picture, $\triangle PQ\Omega$ has one omega-point, $\triangle P\Omega\Theta$ has two, and $\triangle \Omega\Theta\Xi$ three!

Amazingly, many of the standard results of absolute geometry also apply to omega-triangles! The first can be thought of as the AAA congruence theorem where one ‘angle’ is zero.



Theorem 4.21 (Angle-Angle Congruence for Omega-triangles). Suppose $\triangle AB\Omega$ and $\triangle PQ\Theta$ are omega-triangles, each with a single omega-point. If the (finite) angles are congruent in pairs

$$\angle AB\Omega \cong \angle PQ\Theta \quad \angle BA\Omega \cong \angle QP\Theta$$

then the finite sides of the omega-triangles are also congruent: $\overline{AB} \cong \overline{PQ}$.

Remember that omega-points are not really part of hyperbolic geometry—their appearance in our description is a quirk of the Poincaré disk model. It therefore doesn’t make much sense to speak of congruent infinite sides or of congruent angles at omega-points.

Proof. WLOG and for contradiction, assume $\overline{AB} > \overline{PQ}$. Transfer $\angle QP\Theta$ to A to obtain $C \in \overline{AB}$ such that $\overline{AC} \cong \overline{PQ}$. Transferring $\angle PQ\Theta$ creates a ray r based at C on the same side as Ω . Exercise 3 verifies $r = \overline{C\Omega}$. Our hypothesis is that the pictured angles at B and C are congruent.

Let M be the midpoint of $\overline{BC}$ and drop the perpendicular to $R \in \overline{B\Omega}$.

Let $S \in \overline{C\Omega}$ lie on the opposite side of $\overline{BC}$ to R such that $\overline{CS} \cong \overline{BR}$.

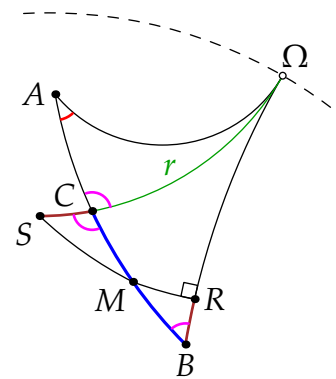
By Side-Angle-Side we have $\triangle MBR \cong \triangle MCS$. In particular:

- $\triangle MCS$ is right-angled(!) at S .
- Congruent vertical angles at M force M to lie on the segment $\overline{RS}$.

The angle of parallelism of S relative to $\overline{B\Omega}$ is therefore $\angle RS\Omega = 90^\circ$, contradicting the Fundamental Theorem (4.15).

There are two other possible orientations:

- R could lie on the opposite side of B from Ω . In this case SAS is applied to the same triangles but with respect to congruent supplementary angles.
- In the special case $R = B$, the same contradiction appears: the angle of parallelism of C with respect to $\overline{B\Omega}$ is 90° .



Theorem 4.22 (Exterior Angle Theorem for Omega-Triangles). Suppose $\triangle BC\Omega$ has a single omega-point and that C lies between A and B . Then $\angle AC\Omega > \angle CB\Omega$.

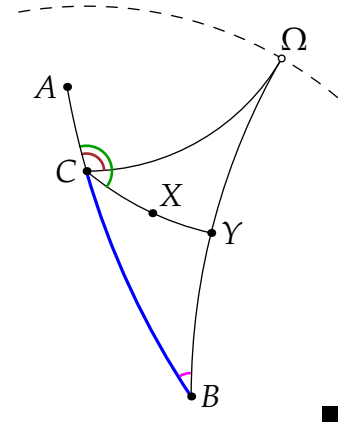
Proof. We show that the two other cases are impossible.

($\angle AC\Omega \cong \angle CB\Omega$) This is precisely the contradictory arrangement described in the previous proof!

($\angle AC\Omega < \angle CB\Omega$) Transfer the **latter** to C to produce $\overrightarrow{CX}$ interior to $\angle BC\Omega$ with $\angle ACX \cong \angle CB\Omega$.

Since $\overrightarrow{C\Omega}$ is a limiting parallel to $\overrightarrow{C\Omega}$, the Fundamental Theorem says that $\overrightarrow{CX}$ intersects $\overrightarrow{B\Omega}$ at some point Y .

But now $\triangle CBY$ contradicts the standard exterior angle theorem ($\angle ACY \cong \angle CBY$).



Corollary 4.23 (Side-Angle Congruence for Omega-triangles). Suppose $\triangle BC\Omega$ and $\triangle PQ\Theta$ both have a single omega-point. If $\angle CB\Omega \cong \angle QP\Theta$ and $\overline{BC} \cong \overline{PQ}$ then $\angle BC\Omega \cong \angle PQ\Theta$.

The final congruence theorem is an exercise based on the previous picture. In summary: A triangle with a single omega-point has three pieces of data (two finite angles and one edge). The AA and SA congruence theorems say that two of these determine the third.

Other observations

Pasch's Axiom & the Crossbar Theorem Versions of these are *theorems* for omega-triangles.

- If a line crosses a side of an omega-triangle and does not pass through any vertex (including Ω), then it must pass through exactly one of the other sides.
- If a line passes through an interior point and exactly one vertex (including Ω) of an omega-triangle, then it passes through the opposite side.

Perpendicular Distance and the Angle of Parallelism Applied to right-angled omega-triangles, the AA and SA theorems prove that the angle of parallelism is a bijective function of the perpendicular distance. Moreover, transferring the right-angle to the positive x -axis and the other vertex to the origin produces the arrangement in Exercise 4.3.6. This completes the proof of Corollary 4.16.

Exercises 4.4. Key concepts: Omega-triangle, AA & SA congruences

1. Let $\triangle PQ\Omega$ be an omega-triangle. Prove that $\angle PQ\Omega + \angle QP\Omega < 180^\circ$.
2. Let ℓ and m be limiting parallels. Explain why they cannot have a common perpendicular.
3. In the proof of the AA congruence, explain why r cannot intersect either $\overrightarrow{A\Omega}$ or $\overrightarrow{B\Omega}$.
4. Prove the Side-Angle congruence theorem for omega-triangles with one omega-point.
5. What would an 'omega-triangle' look like in Euclidean geometry? Comment on the three results in this section: are they still true?

4.5 Area and Angle-defect

In this section we consider one of the triumphs of Johann Lambert: the relationship between the sum of the angles in a triangle and its *area*. We start with a loose axiomatization of area as a relative measure. Until explicitly stated otherwise, we work in *absolute geometry*.

Axiom I Two geometric figures have the same area if and only if they may be sub-divided into finitely many pairs of mutually congruent triangles.³³

Axiom II The area of a triangle is positive.

Axiom III The area of a union of disjoint figures is the sum of the areas of the figures.

Definition 4.24 (Angle defect). Let Σ_{Δ} be the sum of the angles in a triangle. Measured in radians, the *angle-defect* of Δ is $\pi - \Sigma_{\Delta}$.

Since triangles in absolute geometry have $\Sigma_{\Delta} \leq \pi$ (Theorem 4.2), it follows that

$$0 \leq \pi - \Sigma_{\Delta} \leq \pi$$

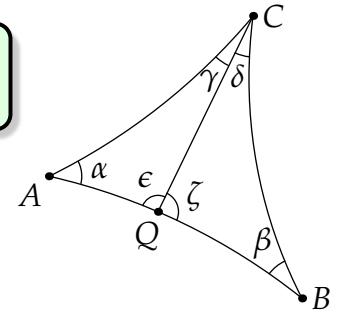
In Euclidean geometry the defect is always zero; in hyperbolic geometry it is strictly positive (Corollary 4.19). A ‘triangle’ with three omega-points would have defect π .

Lemma 4.25. *Angle-defect is additive: If a triangle is split into two sub-triangles, then the defect of the whole is the sum of the defects of the parts.*

This is immediate from the picture:

$$[\pi - (\alpha + \gamma + \epsilon)] + [\pi - (\beta + \delta + \zeta)] = \pi - (\alpha + \beta + \gamma + \delta)$$

since $\epsilon + \zeta = \pi$. Notice that angle-sum is not additive!



Theorem 4.26 (Area determines angle-sum in absolute geometry). *If triangles have the same area, then their angle-sums are identical.*

Of course this is trivial in Euclidean geometry where all triangles have the same angle-sum!

Proof. The lemma provides the induction step: if Δ_1 and Δ_2 have the same area, then their interiors are disjoint unions of a finite collection of mutually congruent triangles:

$$\Delta_1 = \bigcup_{k=1}^n \Delta_{1,k} \quad \text{and} \quad \Delta_2 = \bigcup_{k=1}^n \Delta_{2,k} \quad \text{where} \quad \Delta_{1,k} \cong \Delta_{2,k}$$

Each sub-pair $\Delta_{1,k}, \Delta_{2,k}$ has the same defect, whence the defects of Δ_1 and Δ_2 are equal:

$$\text{defect}(\Delta_1) = \sum_{k=1}^n \text{defect}(\Delta_{1,k}) = \sum_{k=1}^n \text{defect}(\Delta_{2,k}) = \text{defect}(\Delta_2) \quad \blacksquare$$

³³To allow infinitely many infinitesimal sub-triangles would require ideas from analysis...

Angle-sum determines area in hyperbolic geometry

The converse in hyperbolic geometry relies on a beautiful construction generalizing that of Theorem 4.5. The construction is valid in absolute geometry, though the ultimate conclusion that angle-sum determines area is not.

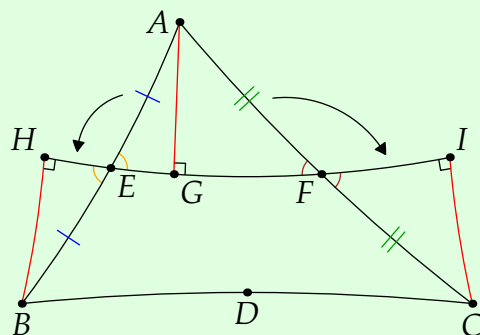
Lemma 4.27. *The picture below illustrates the result of two constructions (2 is the reverse of 1):*

1. Given $\triangle ABC$, bisect sides $\overline{AB}$ and $\overline{AC}$ at E, F respectively and drop **perpendiculars** from the vertices to $\overleftrightarrow{EF}$. Then $\square HICB$ is a Saccheri quadrilateral with summit $\overline{BC}$.
2. Given a Saccheri quadrilateral $\square HICB$ with summit $\overline{BC}$, let A be any point such that $\overleftrightarrow{HI}$ bisects $\overline{AB}$ at E . Then the intersection $F = \overleftrightarrow{HI} \cap \overline{AC}$ is the midpoint of $\overline{AC}$.

Regardless of direction, we have the following conclusions:

- The triangle and quadrilateral have equal area.
- The sum of the summit angles of the quadrilateral equals the angle-sum of the triangle.

We chose $\overline{BC}$ to be the longest side of $\triangle ABC$ —this isn't necessary, though it helpfully forces E, F to lie between H, I .



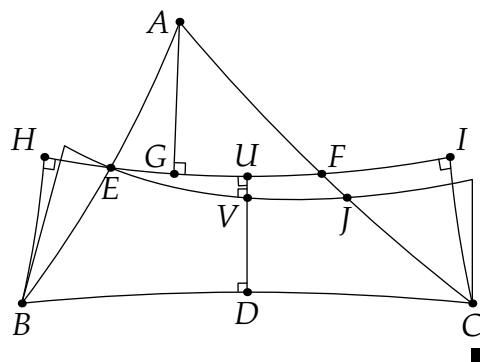
Proof. Part 1 is simply two applications of the SAA congruence theorem (follow the arrows!)

$$\triangle BEH \cong \triangle AEG \text{ and } \triangle CFI \cong \triangle AFG$$

The area and angle-sum correspondence is immediate.

For part 2, suppose the midpoint of $\overline{AC}$ were at $J \neq F$. By part 1, we may create a new Saccheri quadrilateral with summit $\overline{BC}$ using the midpoints E, J .

By Lemma 4.4, the perpendicular bisector of $\overline{BC}$ bisects the bases of both Saccheri quadrilaterals, creating $\triangle EUV$ with two right-angles: contradiction.



We now prove a special case of the main result.

Lemma 4.28. *Suppose **hyperbolic** triangles $\triangle ABC$ and $\triangle PQR$ have congruent sides $\overline{BC} \cong \overline{QR}$ and the same angle-sum. Then the triangles have the same area.*

Proof. Construct the quadrilaterals corresponding to $\triangle ABC$ and $\triangle PQR$ with summits $\overline{BC} \cong \overline{QR}$. These have congruent summits *and* summit angles: by Exercise 4.3.1b they are congruent. ■

The final observation is what makes this special to *hyperbolic* geometry. In the Euclidean case,

Saccheri quadrilaterals are *rectangles*, and congruent summits do not force congruence of the remaining sides.

Theorem 4.29. *In hyperbolic geometry, if $\triangle ABC$ and $\triangle PQR$ have the same angle-sum then they have the same area.*

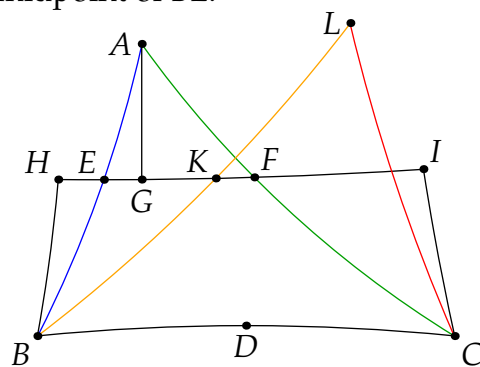
Proof. If the triangles have a pair of congruent edges, the previous result says we are done. Otherwise, use Lemma 4.27 to create a new triangle $\triangle LBC$ which matches the same Saccheri quadrilateral as $\triangle ABC$.

WLOG suppose $|AB| < |PQ|$ and construct the Saccheri quadrilateral with summit $\overline{BC}$. Select K on $\overrightarrow{EF}$ such that $|BK| = \frac{1}{2}|PQ|$ and extend such that K is the midpoint of $\overline{BL}$.

- By Lemma 4.27,

$$\text{Area}(\triangle LBC) = \text{Area}(HICB) = \text{Area}(\triangle ABC)$$

- By Theorem 4.26, $\triangle LBC$ has the same angle-sum as $\triangle ABC$ and thus $\triangle PQR$.
- $\triangle LBC$ and $\triangle PQR$ share a congruent side ($\overline{LB} \cong \overline{PQ}$) and have the same angle-sum. Lemma 4.28 says their areas are equal.



Since both area and angle-*defect* are additive, we immediately conclude:

Corollary 4.30. *The angle-defect of a hyperbolic triangle is an additive function of its area. By normalizing the definition of area,³⁴ we may conclude that*

$$\pi - \Sigma_{\Delta} = \text{Area } \Delta$$

³⁴We have really only proved that $\pi - \Sigma_\Delta$ is proportional to Area Δ . However, it can be seen that these quantities are equal if we use the area measure arising naturally from the hyperbolic distance function (Definition 4.10).

Corollary 4.30 is a special case of the famous Gauss–Bonnet theorem from differential geometry: for any triangle on a surface with Gauss curvature K , we have

$$\Sigma_{\Delta} - \pi = \iint_{\Delta} K \, dA$$

We've now met all three special constant-curvature examples of this:

Euclidean space is *flat* ($K = 0$) so the angle-defect is always zero.

Hyperbolic space has constant negative curvature $K = -1$, whence $\iint_{\Delta} dA = -(\Sigma_{\Delta} - \pi)$ is the angle-defect.

Spherical geometry A radius 1 sphere has *constant positive curvature* $K = 1$ and $\iint_{\Delta} dA$ is the angle-excess $\Sigma_{\Delta} - \pi$.

Note finally how the AAA congruence (Corollary 4.19, part 3) is related to the corollary:

$$\begin{array}{ccc}
 \triangle ABC \cong \triangle DEF & \xLeftrightarrow{\text{AAA}} & \text{angles congruent in pairs} \\
 \downarrow & & \downarrow \\
 \text{equal area} & \xLeftrightarrow{\text{Cor 4.30}} & \text{same angle-defect}
 \end{array}$$

Example (4.11, cont). The isosceles right-triangle with vertices $O, P = (\frac{1}{2}, 0)$ and $Q = (0, \frac{1}{2})$ has angle-sum and area

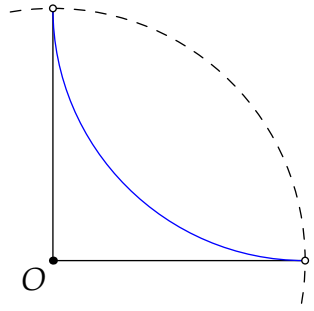
$$\frac{\pi}{2} + 2 \tan^{-1} \frac{3}{5} \approx 151.93^\circ \Rightarrow \text{area} = \pi - \left(\frac{\pi}{2} + 2 \tan^{-1} \frac{3}{5} \right) = \frac{\pi}{2} - 2 \tan^{-1} \frac{3}{5} \approx 0.490$$

A Euclidean triangle with the same vertices has much smaller area: $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} = 0.125$.

Generalizing this (Exercise 4.2.4), the triangle with vertices $O, P = (c, 0)$ and $Q = (0, c)$ has area

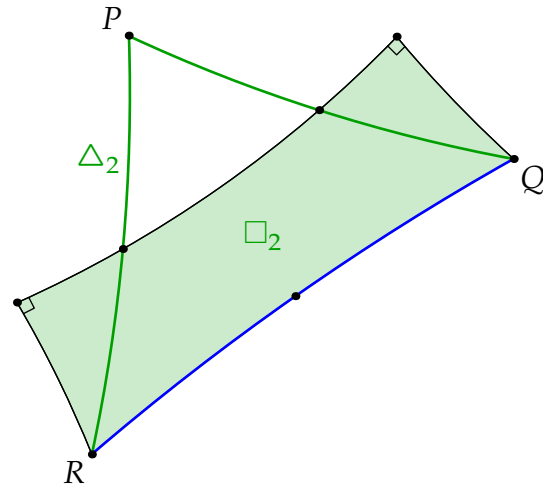
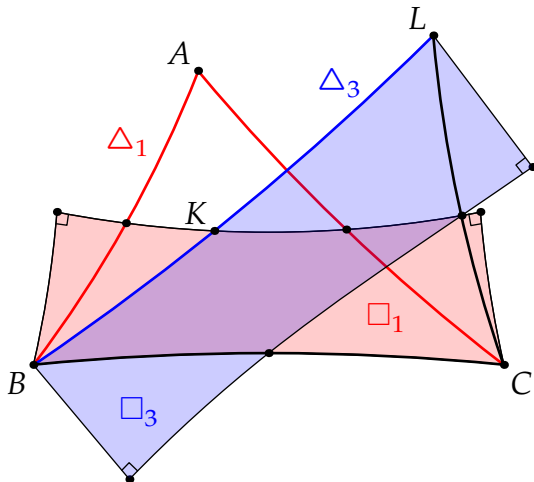
$$\pi - \left(\frac{\pi}{2} + 2 \tan^{-1} \frac{1-c^2}{1+c^2} \right) = \frac{\pi}{2} - 2 \tan^{-1} \frac{1-c^2}{1+c^2}$$

As expected, $\lim_{c \rightarrow 0^+} \text{area}(c) = 0$. In the other limit, the triangle becomes an omega-triangle with two omega-points and $\lim_{c \rightarrow 1^-} \text{area}(c) = \frac{\pi}{2}$: an infinite 'triangle' with finite 'area'!



The limit $c \rightarrow 1^-$

Our discussion in fact provides an explicit method for cutting a triangle into sub-triangles and rearranging its pieces to create a triangle with equal area.



Suppose $\triangle_1$ and $\triangle_2$ have equal areas and construct the quadrilaterals $\square_1$ and $\square_2$. Let L, K be chosen so that $\overline{BL} \cong \overline{QR}$ and K is the midpoint of $\overline{BL}$. We now have:

- $\triangle_1, \triangle_2, \triangle_3, \square_1, \square_2, \square_3$ have the same area.

- The summit angles of $\square_1, \square_2, \square_3$ are congruent (half the angle-sum of each triangle).
- $\square_2, \square_3$ are *congruent* since they have congruent summits and summit angles.

We can now follow the steps in Lemma 4.27 to transform $\triangle_1$ into $\triangle_2$:

$$\triangle_1 \rightarrow \square_1 \rightarrow \triangle_3 \rightarrow \square_3 \cong \square_2 \rightarrow \triangle_2$$

where each arrow represents cutting off two sub-triangles and moving them. Indeed this works even for triangles in Euclidean geometry!

Exercises 4.5. *Key concepts: Area equals angle sum in hyperbolic geometry*

1. Use Corollary 4.30 to find the area of the hyperbolic triangle with given vertices. Your answers to exercises from Section 4.2 should supply the angles!

(a) $O = (0, 0), P = g\left(\frac{1}{2}, \sqrt{\frac{5}{12}}\right)$ and $Q = \left(\frac{1}{2}, -\sqrt{\frac{5}{12}}\right)$.

(b) $O = (0, 0), P = \left(\frac{1}{4}, 0\right), Q = \left(0, \frac{1}{2}\right)$.

(c) $P = (r, 0), Q = \left(-\frac{r}{2}, \frac{\sqrt{3}r}{2}\right), R = \left(-\frac{r}{2}, -\frac{\sqrt{3}r}{2}\right)$ where $0 < r < 1$.

2. Use Lemma 4.27 to give a very cheap proof of part 2 of Corollary 4.19, that the angle-sum in a hyperbolic triangle is strictly less than 180° .
3. In the proof of Theorem 4.29, explain why we can find K such that $|BK| = \frac{1}{2}|PQ|$.
4. Show that there is no *finite* triangle in hyperbolic geometry that achieves the maximum area bound π .
(Hard!) Prove that omega-triangles also satisfy the angle-defect formula: $\text{Area} = \pi - \Sigma_\Delta$, so that only triangles with three omega-points have maximum area.

5. Let $\Omega_1, \dots, \Omega_n$ be n distinct omega-points arranged counter-clockwise around the boundary circle of the Poincaré disk. A region is bounded by the n hyperbolic lines

$$\overrightarrow{\Omega_1\Omega_2}, \overrightarrow{\Omega_2\Omega_3}, \dots, \overrightarrow{\Omega_n\Omega_1}$$

What is the area of the region? Hence argue that the 'area' of hyperbolic space is infinite.

6. An omega-triangle has vertices $O = (0, 0), \Omega = (1, 0)$ and $P = (0, h)$ where $h > 0$.

- (a) Prove that the hyperbolic segment $\overline{P\Omega}$ is an arc of a circle with equation

$$(x - 1)^2 + (y - k)^2 = k^2 \quad (\text{some } k > 0)$$

- (b) Prove that the area of $\triangle OP\Omega$ is given by

$$A(h) = \sin^{-1} \frac{2h}{1 + h^2}$$

7. Suppose two Saccheri quadrilaterals in absolute geometry have the same area and congruent summits. Prove that the quadrilaterals are congruent.

4.6 Isometries and Calculation

There are (at least!) two major issues in our approach to hyperbolic geometry.

Calculations are difficult In (Euclidean) analytic geometry we typically choose the origin and orient axes to ease calculation. We'd like to do the same in hyperbolic geometry.

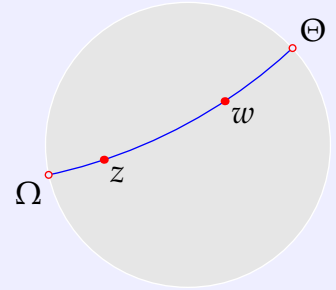
Too many assumptions *Distance, angle* and *line* were defined separately, but in (Euclidean) analytic geometry, these concepts are *not independent*: the distance function defines the other two.³⁵

Isometries provide a related remedy for these issues. To describe these it is helpful to start with an alternative definition of the Poincaré disk and its distance function.

Definition 4.31. The *Poincaré disk* is the set $D := \{z \in \mathbb{C} : |z| < 1\}$ equipped with the distance function

$$d(z, w) := \left| \ln \frac{|z - \Omega| |w - \Theta|}{|z - \Theta| |w - \Omega|} \right|$$

where Ω, Θ are the omega-points for the hyperbolic line through z, w (a diameter, or a circular arc intersecting the boundary circle perpendicularly).



It is not worth memorizing this formula: Corollary 4.38 will show that this is exactly the same as the original cosh formula (Definition 4.10). Indeed it is already easy to check that we get the same formula when $z \neq 0 = w$: since the hyperbolic line $\overleftrightarrow{zw}$ is the diameter of D through z , its omega-points are $\Omega, \Theta = \pm \frac{z}{|z|}$, and we easily compute the distance

$$d(z, 0) = \left| \ln \frac{\left| z - \frac{z}{|z|} \right| \left| 0 + \frac{z}{|z|} \right|}{\left| z + \frac{z}{|z|} \right| \left| 0 - \frac{z}{|z|} \right|} \right| = \ln \frac{1 + |z|}{1 - |z|}$$

exactly as before. Showing that this distance function also determines the usual angle-measure and that paths of shortest distance are indeed hyperbolic lines requires some multivariable calculus.³⁶

³⁵The Pythagorean distance function $|\mathbf{u}| = |PQ| = \sqrt{(q_x - p_x)^2 + (q_y - p_y)^2}$ defines the dot product and therefore the angle measure,

$$|\mathbf{u}| |\mathbf{v}| \cos \theta = \mathbf{u} \cdot \mathbf{v} = \frac{1}{2} (|\mathbf{u} + \mathbf{v}|^2 - |\mathbf{u}|^2 - |\mathbf{v}|^2)$$

With some calculus, the arc-length $\int_C ds = \int_{t_0}^{t_1} |\mathbf{r}'(t)| dt$ of any parametrized curve may be computed. One then shows that the paths of shortest length (*geodesics*) are straight lines.

³⁶A non-examinable discussion of this begins on page ??.

A Little Complex Analysis

For candidate isometries we need functions $f : D \rightarrow D$ for which $d(f(z), f(w)) = d(z, w)$. These follow from some standard results of complex analysis, which we state without proof.

Theorem 4.32 (Möbius/fractional-linear transformations). *If $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$, then the function $f(z) = \frac{az+b}{cz+d}$ has the following properties:*

1. (Invertibility) $f : \mathbb{C} \cup \{\infty\} \rightarrow \mathbb{C} \cup \{\infty\}$ is bijective, with inverse $f^{-1}(z) = \frac{dz-b}{-cz+a}$.
2. (Conformality) If curves intersect, then their images under f intersect at the same angle.
3. (Line/circle preservation) Every line/circle³⁷ in $\mathbb{C}$ is mapped by f to another line/circle.
4. (Cross-ratio preservation) Given distinct z_1, z_2, z_3, z_4 , we have

$$\frac{(f(z_1) - f(z_2))(f(z_3) - f(z_4))}{(f(z_2) - f(z_3))(f(z_4) - f(z_1))} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_2 - z_3)(z_4 - z_1)}$$

The isometries of the Poincaré disk are a subset of the Möbius transformations.

Theorem 4.33. *The orientation-preserving³⁸ isometries of the Poincaré disk have the form*

$$f(z) = e^{i\theta} \frac{\alpha - z}{\bar{\alpha}z - 1} \quad \text{where } |\alpha| < 1 \text{ and } \theta \in [0, 2\pi) \quad (*)$$

All isometries can be found by composing f with complex conjugation (reflection in the real axis).

Referring to the properties in Theorem 4.32:

1. The isometries are precisely the set of Möbius transformations which map omega-points to omega-points and D onto itself (Exercise ??).
2. Isometries preserve angles.
3. The class of hyperbolic lines is preserved: any (Euclidean) circle or line intersecting the unit circle at right-angles is mapped to another such (angle-preservation is used here).
4. If Ω, Θ are the omega-points on $\overleftrightarrow{zw}$, then (by 2 and 3), $f(\Omega)$ and $f(\Theta)$ are the omega-points for the hyperbolic line through $f(z), f(w)$. Preservation of the cross-ratio says that f is an isometry:

$$d(f(z), f(w)) = \left| \ln \frac{|f(z) - f(\Omega)| |f(w) - f(\Theta)|}{|f(z) - f(\Theta)| |f(w) - f(\Omega)|} \right| = \left| \ln \frac{|z - \Omega| |w - \Theta|}{|z - \Theta| |w - \Omega|} \right| = d(z, w)$$

³⁷In $\mathbb{C} \cup \{\infty\}$ a line is just a circle containing ∞ !

³⁸If v lies to the left of $\overleftrightarrow{zw}$, then $f(v)$ is to the left of $\overleftrightarrow{f(z)f(w)}$. This is the usual ‘right-hand rule.’

Computation with Isometries

How does this help us compute? The isometry $(*)$ moves α to the origin, where calculating distances and angles is easy!

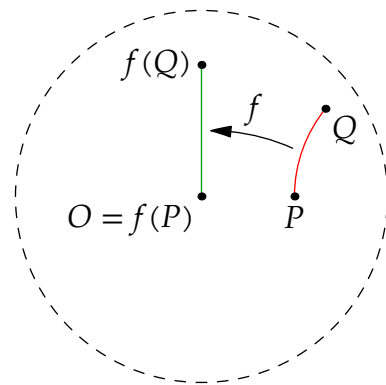
Example 4.34. We compute the hyperbolic distance between the points $P = \frac{1}{2}$ and $Q = \frac{2}{3} + \frac{\sqrt{2}}{3}i$.

Start by moving P to the origin using an isometry with $\alpha = P = \frac{1}{2}$: the natural choice is

$$f(z) = \frac{\alpha - z}{\bar{\alpha}z - 1} = \frac{1 - 2z}{z - 2}$$

Plainly $f(P) = O$. Now compute what f does to Q :

$$f(Q) = \frac{1 - \frac{4}{3} - \frac{2\sqrt{2}}{3}i}{\frac{2}{3} - 2 + \frac{\sqrt{2}}{3}i} = -\frac{1 + 2\sqrt{2}i}{-4 + \sqrt{2}i} = \frac{i}{\sqrt{2}}$$



We could have combined this with a rotation ($e^{i\theta} = e^{-\frac{i\pi}{2}} = -i$) to move $f(Q)$ to the positive x -axis, but there is no need to do so.

Now let us compute distances. By Definition 4.31,

$$d(f(P), f(Q)) = \ln \frac{1 + |f(Q)|}{1 - |f(Q)|} = \ln \frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}} = \ln \frac{\sqrt{2} + 1}{\sqrt{2} - 1} = \ln(3 + 2\sqrt{2})$$

Compare this with the original cosh-formula (Definition 4.10):

$$\begin{aligned} |P|^2 &= \frac{1}{4}, & |Q|^2 &= \frac{4}{9} + \frac{2}{9} = \frac{2}{3}, & |PQ|^2 &= \frac{1}{36} + \frac{2}{9} = \frac{1}{4} \\ d(P, Q) &= \cosh^{-1} \left(1 + \frac{2|PQ|^2}{(1 - |P|^2)(1 - |Q|^2)} \right) = \cosh^{-1} \left(1 + \frac{\frac{2}{4}}{(1 - \frac{1}{4})(1 - \frac{2}{3})} \right) \\ &= \cosh^{-1} 3 = \ln(3 + \sqrt{3^2 - 1}) = \ln(3 + 2\sqrt{2}) \\ &= d(f(P), f(Q)) \end{aligned}$$

If we trust the original cosh-formula, then the transformed points really are the same distance apart as the originals!

It is worth doing some additional verifications (these are exercises):

- The segment $\overline{PQ}$ has equation $(x - \frac{5}{4})^2 + y^2 = \frac{9}{16}$. It can be checked directly that f maps points on this curve to points on the y -axis: the segment $\overline{PQ}$ really is transformed to $\overline{f(P)f(Q)}$, as claimed.
- We could directly compute the co-ordinates of the omega-points for $\overline{PQ}$ and use Definition 4.31 to verify the hyperbolic distance without resorting to the (as yet unsubstantiated) cosh-formula.

Isometries make computing angles *much* easier (recall the horror of Example 4.11...).

Example 4.35. Given $A = -\frac{i}{2}$, $B = -\frac{i}{5}$, $C = \frac{1}{5}(3 - i)$, we find $d(A, B)$, $d(A, C)$ and $\angle A = \angle CBA$. Start by moving A to the origin and consider $f(B)$, $f(C)$:

$$f(z) = \frac{-\frac{i}{2} - z}{\frac{i}{2}z - 1} = \frac{2z + i}{2 - iz} \implies f(B) = \frac{-\frac{2i}{5} + i}{2 - \frac{1}{5}} = \frac{i}{3}$$

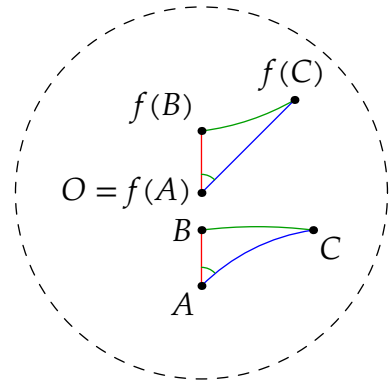
$$f(C) = \frac{\frac{2}{5}(3 - i) + i}{2 - \frac{i}{5}(3 - i)} = \frac{2(3 - i) + 5i}{10 - i(3 - i)} = \frac{2 + i}{3 - i} = \frac{(2 + i)(3 + i)}{10} = \frac{1 + i}{2}$$

Since $f(A)$ is the origin, **two sides** of the triangle are now *Euclidean straight lines* and the computations become easy:

$$d(A, B) = d(O, f(B)) = \ln \frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} = \ln 2$$

$$d(A, C) = d(O, f(C)) = \ln \frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}} = 2 \ln(\sqrt{2} + 1)$$

$$\angle A = \angle f(C)f(A)f(B) = \arg \frac{i}{3} - \arg \frac{1 + i}{2} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$



To compute the **final side** and angles, we could instead move B and then C to the origin.

Interpretation of Isometries (non-examinable) As in Euclidean geometry, isometries can be interpreted as rotations, reflections and translations. Here is the dictionary.

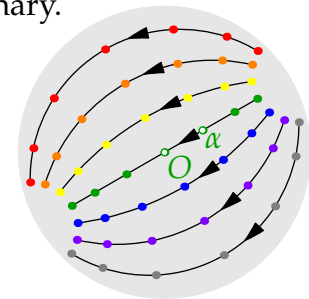
Translations $T_{-\alpha}(z) = \frac{\alpha - z}{\bar{\alpha}z - 1}$ translates α to the origin. The first picture shows repeated applications of $T_{-\alpha}$ to seven initial points. Compose such maps to translate α to β :

$$T_{\beta} \circ T_{-\alpha}(z) = \frac{(\bar{\alpha}\beta - 1)z + \alpha - \beta}{(\bar{\alpha} - \bar{\beta})z + \alpha\bar{\beta} - 1}$$

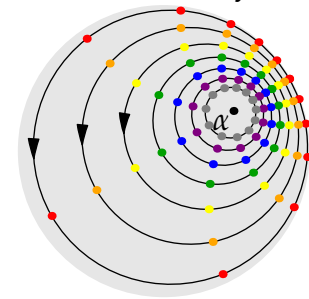
Rotations $R_{\theta}(z) = e^{i\theta}z$ rotates counter-clockwise around the origin. To rotate around α , use the composition

$$T_{\alpha} \circ R_{\theta} \circ T_{-\alpha}$$

Reflections $P_{\theta}(z) = e^{2i\theta}\bar{z}$ reflects across the line making angle θ with the positive real axis. Composition permits more general reflections, e.g., $T_{\alpha} \circ P_{\theta} \circ T_{-\alpha}$ reflects across the line through α making angle θ with the positive real direction.



Translation by α



Rotation by $\frac{\pi}{6}$ around α

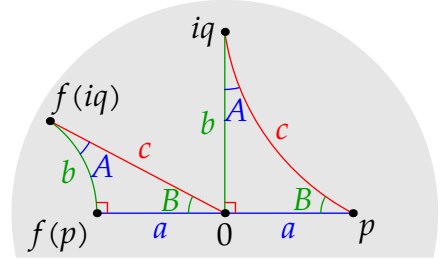
Hyperbolic Trigonometry

By employing isometries in the abstract, we develop expressions that allow us to solve hyperbolic triangles directly using only its side-lengths and angle-measures.

Start with a right-triangle. We may suppose an isometry has already moved the right-angle to the origin and the other sides to the positive axes at p, iq as pictured. The non-hypotenuse hyperbolic side-lengths are therefore

$$a = \ln \frac{1+p}{1-p} = \cosh^{-1} \frac{1+p^2}{1-p^2}, \quad b = \cosh^{-1} \frac{1+q^2}{1-q^2}$$

This uses only Definition 4.31 and the definition of inverse-cosh (footnote 30). To measure the hypotenuse, translate p to the origin via an isometry



$$f(z) = \frac{p-z}{pz-1} \Rightarrow f(iq) = \frac{p-iq}{ipq-1} \Rightarrow |f(iq)|^2 = \frac{p^2+q^2}{p^2q^2+1}$$

We therefore see that

$$\cosh c = \frac{1+|f(iq)|^2}{1-|f(iq)|^2} = \frac{1+p^2q^2+p^2+q^2}{1+p^2q^2-p^2-q^2} = \frac{1+p^2}{1-p^2} \cdot \frac{1+q^2}{1-q^2} = \cosh a \cosh b$$

Applying the hyperbolic identity $\sinh^2 b = \cosh^2 b - 1$, we obtain

$$\sinh b = \frac{2q}{1-q^2} \Rightarrow \tanh b = \frac{\sinh b}{\cosh b} = \frac{2q}{1+q^2}$$

Writing $f(iq)$ in real and imaginary parts allows us to find the **slope** (that is, $\tan B$):

$$f(iq) = \frac{p-iq}{ipq-1} = \frac{-p(1+q^2)+iq(1-p^2)}{p^2q^2+1} \Rightarrow \tan B = \frac{q(1-p^2)}{p(1+q^2)} = \frac{\tanh b}{\sinh a}$$

Trigonometric identities such as $\csc^2 B = 1 + \cot^2 B$ quickly yield the other functions.

Theorem 4.36. *In a hyperbolic right-triangle with adjacent a , opposite b , and hypotenuse c ,*

$$\sin B = \frac{\sinh b}{\sinh c} \quad \cos B = \frac{\tanh a}{\tanh c} \quad \tan B = \frac{\tanh b}{\sinh a} \quad \cosh c = \cosh a \cosh b$$

This last is Pythagoras' Theorem for hyperbolic triangles.

Pythagoras is easy to remember, as are the $\frac{\text{opp}}{\text{hyp}}, \frac{\text{adj}}{\text{hyp}}, \frac{\text{opp}}{\text{adj}}$ patterns for the basic trig functions. Otherwise, these expressions (and the upcoming Corollary 4.38) are open-book—they are not worth memorizing.

Examples 4.37. The basic rules of hyperbolic trigonometry allow us to analyze right-triangles just as in Euclidean geometry.

1. A right-triangle has $a = \cosh^{-1} 3 \approx 1.76$ and $b = \cosh^{-1} 5 \approx 2.29$ (these are the triangles pictured in the previous analysis). We compute the remaining sides and angles.

First apply Pythagoras to compute the hypotenuse:

$$\cosh c = \cosh a \cosh b = 15 \Rightarrow c = \cosh^{-1} 15 \approx 3.40$$

Now use the formula for sine to find the two non-right angles:

$$\sin A = \frac{\sinh a}{\sinh c} = \sqrt{\frac{\cosh^2 a - 1}{\cosh^2 c - 1}} = \sqrt{\frac{3^2 - 1}{15^2 - 1}} = \frac{1}{2\sqrt{7}} \Rightarrow A \approx 10.9^\circ$$

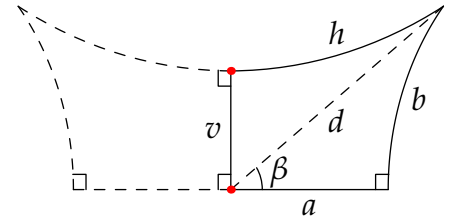
$$\sin B = \frac{\sinh b}{\sinh c} = \sqrt{\frac{\cosh^2 b - 1}{\cosh^2 c - 1}} = \sqrt{\frac{5^2 - 1}{15^2 - 1}} = \sqrt{\frac{3}{28}} \Rightarrow B \approx 19.1^\circ$$

We could instead have used the other trigonometric expressions: for instance,

$$\tan A = \frac{\tanh a}{\sinh b} = \frac{\sinh a}{\cosh a \sinh b} = \frac{\sqrt{8}}{3\sqrt{24}} = \frac{1}{3\sqrt{3}} \Rightarrow A = \tan^{-1} \frac{1}{3\sqrt{3}} \approx A \approx 10.9^\circ$$

2. Consider the pictured Lambert quadrilateral with side-lengths a, b, v, h and diagonal d . Applying the sine, cosine and tangent formulae,

$$\begin{aligned} \sin \beta &= \frac{\sinh b}{\sinh d} & \cos \beta &= \sin \left(\frac{\pi}{2} - \beta \right) = \frac{\sinh h}{\sinh d} \\ \Rightarrow \frac{\tanh b}{\sinh a} &= \tan \beta = \frac{\sin \beta}{\cos \beta} = \frac{\sinh b}{\sinh h} \\ \Rightarrow \sinh h &= \sinh a \cosh b \end{aligned}$$



By doubling the quadrilateral, we obtain a Saccheri quadrilateral with base $2a$, congruent sides b , and summit $2h$. Since $\cosh b > 1$ and is strictly increasing, observe:

- The summit of a Saccheri quadrilateral is *longer* than its base.
- If the base $2a$ is fixed, the summit is a *strictly increasing* function of the side-length b .

If such a line exists, a common perpendicular describes the shortest path between two parallel lines.

The goal of trigonometry is to 'solve' triangles: given minimal numerical data, to compute the remaining sides and angles. As in Euclidean geometry, we can drop perpendiculars and using the results of Theorem 4.36. Alternatively, we can generalize this procedure by developing the sine and cosine rules.

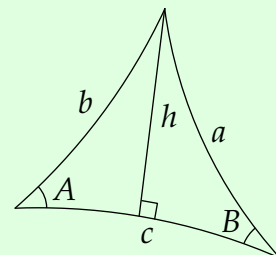
Corollary 4.38 (Sine/Cosine rules and the Cosh-distance formula). Label a general triangle with angle-measures A, B, C opposite sides with (hyperbolic) lengths a, b, c .

Sine Rule This is proved exactly as the easy method in Euclidean geometry.

Drop a perpendicular from C to obtain $\sin A = \frac{\sinh h}{\sinh b}$ and $\sin B = \frac{\sinh h}{\sinh a}$.
Eliminate $\sinh h$ for the first equality in

$$\frac{\sinh a}{\sin A} = \frac{\sinh b}{\sin B} = \frac{\sinh c}{\sin C}$$

Drop a different altitude for the remaining equality.



Cosine Rule I Repeat the argument of Theorem 4.36 for a triangle with vertices $0, p$ and qe^{iC} to obtain

$$\cosh c = \cosh a \cosh b - \sinh a \sinh b \cos C$$

Expressing the right side in terms of p, q ($\cosh a = \frac{1+p^2}{1-p^2}$, etc.) yields the original cosh-formula for distance (Definition 4.10).

Cosine Rule II Hyperbolic geometry admits a second version (for a proof see Exercise ??),

$$\cos C = \sin A \sin B \cosh c - \cos A \cos B$$

In hyperbolic geometry, the triangle congruence theorems (SAS, ASA, SSS, SAA and AAA) provide suitable minimal data. The second version of the cosine rule has no analogue in Euclidean geometry—it is particularly helpful for solving triangles given ASA or AAA data.³⁹

Examples 4.39. We give four examples of solving hyperbolic triangles.

1. (SAS) An isosceles triangle has $C = \frac{\pi}{3}$ and $a = b = \cosh^{-1} 2 \approx 1.32$. We have

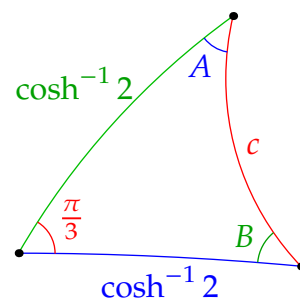
$$\sinh a = \sinh b = \sqrt{\cosh^2 a - 1} = \sqrt{3}$$

By the sine and cosine rules,

$$\cosh c = 2 \cdot 2 - \sqrt{3}\sqrt{3} \cdot \frac{1}{2} = \frac{5}{2} \Rightarrow c = \cosh^{-1} \frac{5}{2} \approx 1.57$$

$$\sin B = \sin A = \frac{\sin C \sinh a}{\sinh c} = \frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{21}}{4}} = \frac{3}{\sqrt{21}} = \sqrt{\frac{3}{7}}$$

$$\Rightarrow A = B \approx 40.9^\circ$$

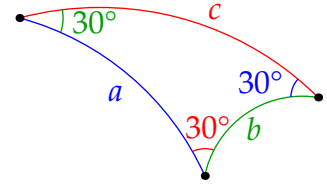


³⁹Unlike in Euclidean geometry, knowing two angles doesn't automatically give you the third! For SAS and SSS start with the cosine rule. SAA data is best solved by dropping a perpendicular and using Theorem 4.36.

2. (Equilateral AAA) An equilateral triangle has interior angles 30° . Applying the second cosine rule, we see that

$$\cosh c = \frac{\cos A \cos B + \cos C}{\sin A \sin B} = \frac{\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}}{\frac{1}{2} \cdot \frac{1}{2}} = 3\sqrt{3}$$

$$\Rightarrow a = b = c = \cosh^{-1}(3\sqrt{3}) \approx 2.33$$



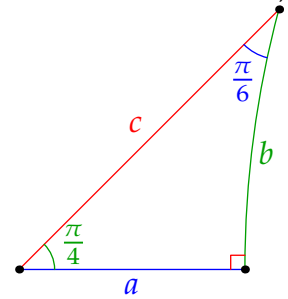
3. (Right-angled AAA) A triangle has angles $\frac{\pi}{6}$, $\frac{\pi}{4}$ and $\frac{\pi}{2}$. Rather than using the second version of the cosine rule, we indicate part of its proof by employing the tan-formula twice,

$$\frac{1}{\sqrt{3}} = \tan \frac{\pi}{6} = \frac{\tanh a}{\sinh b} = \frac{\sinh a}{\cosh a \sinh b}$$

$$1 = \tan \frac{\pi}{4} = \frac{\tanh b}{\sinh a} = \frac{\sinh b}{\sinh a \cosh b}$$

Multiply these together and apply hyperbolic Pythagoras,

$$\frac{1}{\sqrt{3}} = \frac{1}{\cosh a \cosh b} = \frac{1}{\cosh c} \Rightarrow c = \cosh^{-1} \sqrt{3} = \ln(\sqrt{3} + \sqrt{2}) \approx 1.15$$



Since $\sinh c = \sqrt{\cosh^2 c - 1} = \sqrt{2}$, the remaining sides are easy to compute:

$$\frac{1}{\sqrt{2}} = \sin \frac{\pi}{4} = \frac{\sinh b}{\sinh c} \Rightarrow b = \sinh^{-1} 1 = \cosh^{-1} \sqrt{2} \approx 0.88$$

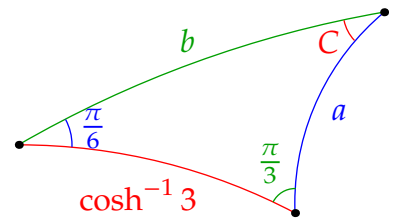
$$\cosh a = \frac{\cosh c}{\cosh b} = \sqrt{\frac{3}{2}} \Rightarrow a = \cosh^{-1} \sqrt{\frac{3}{2}} = \sinh^{-1} \frac{1}{\sqrt{2}} \approx 0.66$$

4. (ASA) Solve a triangle with angles $\frac{\pi}{6}$, $\frac{\pi}{3}$ and a distance $\cosh^{-1} 3$ between them.

Apply the second version of the cosine rule:

$$\cos C = \sin A \sin B \cosh c - \cos A \cos B$$

$$= \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \cdot 3 - \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = \frac{\sqrt{3}}{2} \Rightarrow C = \frac{\pi}{6}$$



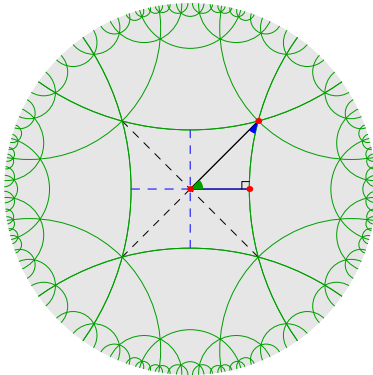
The triangle is isosceles, whence $a = \cosh^{-1} 3 \approx 1.76$ also. By the cosine rule,

$$\cosh b = \cosh a \cosh c - \sinh a \sinh c \cos B = 9 - (\sqrt{3^2 - 1})^2 \frac{1}{2} = 5$$

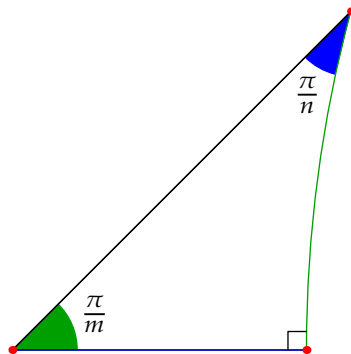
$$\Rightarrow b = \cosh^{-1} 5 = \ln(5 + \sqrt{24}) \approx 2.29$$

Hyperbolic Tilings (just for fun!)

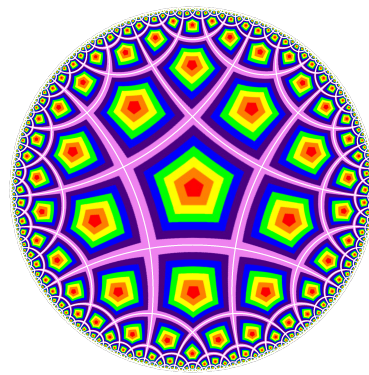
Example 4.39.3 can be used to construct regular tilings of hyperbolic space. Take eight congruent copies of the triangle and arrange them around the origin as in the first picture. Now reflect the quadrilateral over each of its edges and repeat the process in all directions ad infinitum. The result is a regular tiling of hyperbolic space comprising *four-sided* figures with *six* meeting at every vertex!



$(m, n) = (4, 6)$



The fundamental triangle



$(m, n) = (5, 4)$

In hyperbolic space, many different regular tilings are possible. Suppose such is to be made using regular m -sided polygons, n of which are to meet at each vertex: each polygon comprises $2m$ copies of the pictured *fundamental right-triangle*, whose angles must be $\frac{\pi}{2}$, $\frac{\pi}{m}$ and $\frac{\pi}{n}$. Since the angles sum to less than π radians, we see that there exists a regular tiling of hyperbolic space precisely when

$$\frac{\pi}{2} + \frac{\pi}{m} + \frac{\pi}{n} < \pi \iff (m-2)(n-2) > 4$$

The first example is $m = 4$ and $n = 6$. In the other picture tiling, $n = 4$ pentagons ($m = 5$) meet at each vertex (the interior of each pentagon has been colored congruently for fun). This pentagonal tiling was produced using the tools found here and here: have a play!

The multitude of possible tilings in hyperbolic geometry is in contrast to Euclidean geometry, where a regular tiling requires *equality*

$$(m-2)(n-2) = 4$$

The three solutions $(m, n) = (3, 6)$, $(4, 4)$, $(6, 3)$ correspond to the only tilings of Euclidean geometry by regular polygons (equilateral triangles, squares and regular hexagons); unlike in hyperbolic geometry, Euclidean tilings may have arbitrary side-lengths.

For related fun, look up M.C. Escher's *Circle Limit* artworks, some of which are based on hyperbolic tilings. If you want an excuse to play video games while pretending to study geometry, have a look at Hyper Rogue, which relies on (sometimes irregular) tilings of hyperbolic space.

Exercises 4.6. *Key concepts:*

The questions marked * require abstract calculations with complex algebra. Feel free to skip these if your previous experience with this is minimal.

1. Complete the additional verifications in Example 4.34
2. Use Definition 4.31 to prove that $d(z, 0) = \ln \frac{1+|z|}{1-|z|}$.
(Hint: what are the omega-points for the line through 0 and z ?)
3. (a) Use an isometry to find angle $\angle ABC$ when $A = 0$, $B = \frac{i}{2}$, and $C = \frac{1+i}{2}$.
(b) Now compute $\angle ACB$, and thus find the angle-sum and area of the triangle.
4. Find the area of each triangle in Examples 4.39.
5. * Identify a Möbius transformation $f(z) = \frac{az+b}{cz+d}$ with the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If g is another Möbius transformation, prove that the composition $f \circ g$ corresponds to the product of the matrices related to f, g . Verify that $f^{-1}(z) = \frac{dz-b}{a-cz}$ corresponds to the inverse matrix.⁴⁰
6. (a) A triangle has vertices $A = \frac{1}{3}$, $B = \frac{1}{2}$ and C , where $\angle BAC = 45^\circ$ and $b = d(A, C) = \cosh^{-1} 3$. Compute $a = d(B, C)$ using the hyperbolic cosine rule.
(b) The isometry $f(z) = \frac{1-3z}{z-3}$ moves A to the origin. What is $f(B)$ and therefore $f(C)$?
(Hint: remember that f is orientation preserving)
(c) Use the *inverse* of the isometry f to compute the co-ordinates of C . As a sanity-check, use the cosh distance formula to recover your answer to part (a).
7. * Suppose $f(z) = \frac{\alpha-z}{\bar{\alpha}z-1}$ for some constant $\alpha \in \mathbb{C}$ with $|\alpha| \neq 1$. If $|z| = 1$, prove that $|f(z)| = 1$. Argue that the functions f in Theorem 4.33 really do map the interior of the unit disk to itself.
Suppose $f(z) = \frac{az+b}{cz+d}$ maps D bijectively onto itself and that it restricts to a map $f : S^1 \rightarrow S^1$. Suppose that $\alpha \in D$ is the unique value such that $f(\alpha) = 0$.
(a) Show that, WLOG, we may write $f(z) = \frac{z-\alpha}{cz+d}$ where $c\alpha \neq d$.
(b) For $f(z)$ as in part (a), evaluate

$$1 = |f(e^{i\psi})|^2 = \frac{|e^{i\psi} - \alpha|^2}{|ce^{i\psi} + d|^2}$$

Hence conclude that $|c| = 1$ or $|d| = 1$.

- (c) By writing $c = e^{i\theta}$, show that $|c| = 1$ is a contradiction. Hence conclude that $d = e^{-i\theta}$, whence $f(z) = e^{i\theta} \frac{\alpha-z}{\bar{\alpha}-z}$ as claimed.
- (d) Finally, evaluate $|f(re^{i\psi})|$ whenever $0 \leq r < 1$, and thus conclude that such f really do map $f : D \rightarrow D$.

8. (a) * Show that the isometry $T_\beta \circ T_{-\alpha}$ which translates α to β (page 90) is the translation $T_{-\gamma}$ where $\gamma = \frac{\beta - \alpha}{\alpha\beta - 1}$ followed by a rotation around the origin.
 (b) * In what rare situations is the composition of two translations another (pure) translation?
9. Use the power series $\cosh x = 1 + \frac{1}{2}x^2 + \frac{1}{4!}x^4 + \dots$ to expand the hyperbolic Pythagorean theorem $\cosh c = \cosh a \cosh b$ to order 4 (a^4, a^2b^2 , etc.). What do you observe?
10. A hyperbolic right-triangle has non-hypotenuse sides $a = \cosh^{-1} 2$ and $b = \cosh^{-1} 3$. Find the hypotenuse, the angles and the area of the triangle.
11. Given ASA data $c = \cosh^{-1}(\sqrt{2} + \sqrt{3})$, $A = \frac{\pi}{4}$, $B = \frac{\pi}{6}$, find the remaining data for the triangle.
12. An equilateral hyperbolic triangle has side-length a and angle A . Prove that $\cosh a = \frac{\cos A}{1 - \cos A}$. If $A = 45^\circ$, what is the side-length?
13. Find the interior angles and side-lengths for the quadrilateral and pentagonal tiles on page 95.
14. A railway comprises two rails (lines) which start perpendicular to a common sleeper (cross-beam). Why would it be difficult to build a railway in hyperbolic geometry?
 (Hint: consider Example 4.37.2)
15. * As suggested in Corollary 4.38, prove both the cosine rule and the cosh distance formula.
16. You are given isosceles ASA data: angles $A = B$ and side c . Prove that $\cosh c \leq 2 \csc^2 A - 1$. What happens when this is equality?
17. (a) Prove the second cosine rule when $C = \frac{\pi}{2}$ (see the trick in Example 4.39.3).
 (b) (Hard!) Prove the full version by dropping a perpendicular from $B = B_1 + B_2$ and observing that $\frac{\cos A}{\sin B_1} = \frac{\cos C}{\sin B_2} = \frac{\cos C}{\sin(B - B_1)} \dots$

⁴⁰Since multiplying a, b, c, d by a non-zero scalar doesn't change f , the set of Möbius transformations is isomorphic to the *projective special linear group* $\text{PSL}_2(\mathbb{R})$. The orientation-preserving isometries of hyperbolic space form a proper subgroup.

The Poincaré Disk for Differential Geometers (non-examinable)

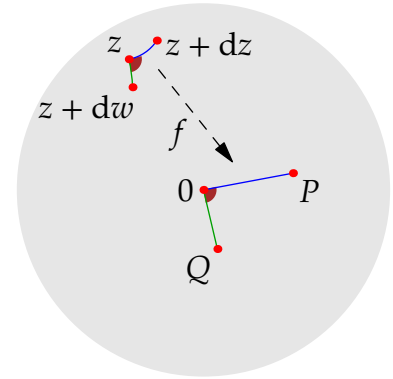
Most of this last optional section should be accessible to anyone who's taken basic vector-calculus. All we really need is the Poincaré disk model with its distance function $d(z, w)$ and a description of the isometries (Theorems 4.32, 4.33).

Consider the infinitesimally separated points z and $z + dz$. Map z to the origin via an isometry

$$f : \xi \mapsto \frac{z - \xi}{\bar{z}\xi - 1}$$

Then $z + dz$ is mapped to

$$P := f(z + dz) = \frac{-dz}{\bar{z}(z + dz) - 1} = \frac{dz}{1 - |z|^2}$$



where we deleted $\bar{z} dz$ since it is infinitesimal compared to $1 - |z|^2$.

Since isometries preserve length and angle, this construction has several consequences.

Infinitesimal distance, arc-length, and geodesics

The hyperbolic distance from z to $z + dz$ is

$$d(z, z + dz) = d(O, P) = \ln \frac{1 + |P|}{1 - |P|} = \ln(1 + |P|) - \ln(1 - |P|) = 2|P| = \frac{2|dz|}{1 - |z(t)|^2} \quad (*)$$

where the approximation $\ln(1 \pm |P|) = \pm |P|$ is used since $|P|$ is infinitesimal. If $z(t)$ parametrizes a curve in the disk, then the infinitesimal distance formula allows us to compute its arc-length

$$\int_{t_0}^{t_1} \frac{2|z'(t)|}{1 - |z(t)|^2} dt$$

Example 4.40 (Circles and 'hyperbolic π '). Suppose that a circle has hyperbolic radius δ . By moving its center to the origin via an isometry, we may parametrize in the usual manner:

$$z(t) = r \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \theta \in [0, 2\pi) \quad \text{where} \quad \delta = \ln \frac{1+r}{1-r} \quad \text{equivalently} \quad r = \frac{e^\delta - 1}{e^\delta + 1}$$

Its circumference (hyperbolic arc-length) is then

$$\int_0^{2\pi} \frac{2r}{1 - r^2} d\theta = \frac{4\pi r}{1 - r^2} = 2\pi \sinh \delta = 2\pi \left(\delta + \frac{1}{3!}\delta^3 + \frac{1}{4!}\delta^5 + \dots \right) > 2\pi\delta$$

where we used the Maclaurin series to compare.

A hyperbolic circle has a *larger* circumference : diameter ratio than for a Euclidean circle (π).

Moreover, this ratio is *not constant*: one might say that the hyperbolic version of π is a function $(\frac{\pi \sinh \delta}{\delta})!$

Our arc-length integral approach also allows us to show that hyperbolic lines are really what we want them to be: lines of shortest distance between points.

Theorem 4.41. *The geodesics—paths of minimal length between two points—in the Poincaré disk are precisely the hyperbolic lines.*

Following the comments on page 87, the distance function really does define the concept of hyperbolic line.

Proof. First suppose b lies on the positive x -axis. Parametrize a curve from 0 to b via

$$z(t) = x(t) + iy(t) \quad \text{where} \quad 0 \leq t \leq 1, \quad z(0) = 0, \quad z(1) = b$$

Its arc-length satisfies

$$\begin{aligned} \int_0^1 \frac{2|z'(t)|}{1-|z(t)|^2} dt &= \int_0^1 \frac{2\sqrt{x'^2 + y'^2}}{1-x^2-y^2} dt \geq \int_0^1 \frac{2|x'|}{1-x^2} dt \geq \int_0^1 \frac{2x'(t)}{1-x(t)^2} dt = \int_0^b \frac{2dx}{1-x^2} \\ &= \ln \frac{1+b}{1-b} = d(0, b) \end{aligned}$$

where we have equality if and only if $y(t) \equiv 0$ and $x(t)$ is increasing. The length-minimizing path is therefore along the x -axis.

More generally, given points A, B , apply an isometry f such that $f(A) = 0$ and $f(B) = b$ lies on the positive x -axis. The geodesic from A to B is therefore the image of the segment $\overline{0b}$ under the inverse isometry f^{-1} . By the properties of Möbius transforms, this is an arc of a Euclidean circle through A, B intersecting the unit circle at right-angles, our original definition of a hyperbolic line. ■

Area Computation

If dx and idy are infinitesimal horizontal and vertical changes in $z = x + iy$, then the area of the infinitesimal rectangle spanned by $z \rightarrow z + dx$ and $z \rightarrow z + idy$ is the area element

$$dA = \frac{2dx}{1-|z|^2} \frac{2dy}{1-|z|^2} = \frac{4dx dy}{(1-x^2-y^2)^2}$$

The area of a region R in the Poincaré disk is therefore given by the double integral

$$\iint_R \frac{4dx dy}{(1-x^2-y^2)^2} = \iint_R \frac{4r dr d\theta}{(1-r^2)^2} = \iint_R \sinh \delta d\delta d\theta$$

where the last expression is written in polar co-ordinates using the hyperbolic distance δ . In this way the measure of area also depends on the distance function.

Example (??, cont). The area of a hyperbolic circle with hyperbolic radius δ is

$$\int_0^{2\pi} \int_0^\delta \sinh \delta \, d\delta \, d\theta = 2\pi(\cosh \delta - 1) = \pi \left(\delta^2 + \frac{2}{4!}\delta^4 + \frac{2}{6!}\delta^6 + \dots \right) > \pi\delta^2$$

Again, this is larger than you'd expect in Euclidean geometry.

Angle Measure and the First Fundamental Form

If we repeat the distance translation (*, page ??) for a second infinitesimal **segment** $z \rightarrow z + dw$, it can be checked that the **angle** between the original segments is precisely that between the infinitesimal vectors dz and dw . This is precisely the conformality observation in Theorem 4.32 and moreover shows how the distance function determines the angle measure.

If you've studied differential geometry, then a more formal way to think about this is to use the *first fundamental form* or *metric*: essentially the dot product of infinitesimal tangent vectors. For the Poincaré disk model, (*) says that this is

$$I = \frac{4(dx^2 + dy^2)}{(1 - x^2 - y^2)^2} = \frac{4(dr^2 + r^2 d\theta^2)}{(1 - r^2)^2}$$

Since this is a scalar multiple of the standard Euclidean metric $dx^2 + dy^2$, angle measures are identical.⁴¹ It also gels with the fact that arc-length is the integral

$$\int \sqrt{I(z'(t), z'(t))} \, dt$$

Using this language, two of the major theorems of introductory differential geometry quickly put a couple of remaining issues to bed.

Gauss' Theorem Egregium The first fundamental form determines the *Gaussian curvature* K . In this case $K = -1$ is constant and negative, as you should easily be able to verify if you've studied differential geometry.

Gauss–Bonnet Theorem The angle-sum Σ_Δ of a geodesic triangle in a space with Gaussian curvature K satisfies

$$\Sigma_\Delta - \pi = \iint_\Delta K$$

This establishes our earlier assertion that $\text{Area } \Delta = \pi - \Sigma_\Delta$ (Corollary 4.30).

⁴¹Recall that the angle ψ between vectors $\mathbf{u}, \mathbf{v}$ satisfies $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos \psi$. For infinitesimal vectors we use $I = \lambda^2(dx^2 + dy^2)$ instead of the dot product, where $\lambda = \frac{2}{1-r^2}$. The resulting angle is the same as if we use the Euclidean metric $dx^2 + dy^2$, since factors of λ^2 cancel on both sides.